분석 보고서

Executive Summary

이제 최종 보고서를 완성하고 HTML과 Markdown 버전을 모두 생성하겠습니다.

연구 데이터 및 배경 정보

No topic

카테고리: No category

수집 시간: No time information

리튬 산업 전반의 감산 및 폐쇄 동향 (2023-2024)

카테고리: industry_trends

수집 시간: 2025-08-07 16:10:44

No topic

카테고리: No category

수집 시간: No time information

리튬 산업 전반의 감산 및 폐쇄 동향 (2023-2024)

시장 상황 개요

2023년부터 2024년까지 리튬 가격은 약 90%가량 하락하였으며, 이는 공급 과잉과 예상보다 낮은 전기차(EV) 수요로 인한 것입니다. 이러한 가격 하락으로 인해 많은 리튬 광산들이 생산량을 줄이거나 운영을 중단하는 상황이 발생했습니다.

주요 감산 및 폐쇄 사례

1. **생산량 감축 규모**:
   - 2023년 말부터 약 190,000톤의 리튬 광산 생산능력이 감축되었으며, 약 50,000톤의 프로젝트가 지연되었습니다. (출처: CRU, Bloomberg 2024년 11월 14일 보도)
   - 호주의 리튬 광산들은 2024년 1분기에 전체적으로 약 12.3%의 생산량 감소를 기록했습니다. (출처: Antaike 통계, lithium-chemical.com)
2. **주요 기업들의 대응**:
   - **Albemarle**: 2024년 1월, 미국 최대 리튬 생산업체인 Albemarle은 인력 감축과 미국 리튬 정제소 프로젝트에 대한 지출 연기를 발표했습니다. (출처: Reuters, 2024년 1월 17일)
   - **Core Lithium**: 호주의 Core Lithium은 2024년 1월 Finniss 리튬 프로젝트의 Grants 광산 채굴을 중단했으며, 2024년 중반에는 가공 시설도 중단하고 모든 인프라를 관리 상태로 전환했습니다. (출처: Core Lithium 공식 웹사이트, Mining Technology)
   - **Pilbara Minerals**: 2024년 12월 Ngungaju 공장을 관리 상태로 전환하기로 결정했으며, 이를 통해 약 2억 호주달러(1억 3,120만 달러)의 현금 흐름 개선을 예상하고 있습니다. (출처: Fastmarkets, 2024년 10월 30일)
3. **중국 생산업체들의 동향**:
   - 여러 중국 리튬 생산업체들이 손실을 줄이기 위해 생산량 감축이나 완전한 중단을 고려하고 있습니다. (출처: Fastmarkets)
   - 2024년 9월, 중국의 한 주요 생산업체가 국내 최대 레피돌라이트(lepidolite) 광산 중 하나의 생산을 중단했으며, 이로 인해 3개의 관련 가공 공장이 폐쇄되었습니다. (출처: Fastmarkets)

시장 전망

1. **공급 과잉 완화 전망**:
   - 2025년에는 리튬 공급 과잉이 절반으로 줄어들 것으로 예상됩니다. (출처: Antaike, Reuters 2025년 1월 13일)
   - 중국의 전기차 보조금이 2025년 가격 지원에 도움이 될 것으로 예상됩니다. (출처: Reuters)
2. **가격 안정화 요인**:

    - 광산 폐쇄와 중국의 전기차 판매 증가로 인해 2025년에는 리튬 가격이 안정화될 것으로 분석가들은 예측하고 있습니다. (출처: Reuters)
    - 그러나 공급망에 상당한 재고가 있어 가격의 강한 회복은 예상되지 않습니다. (출처: Fastmarkets)
3. **장기 전망**:
    - 일부 분석가들은 리튬 과잉 공급이 2028년까지 지속될 수 있다고 예측하고 있습니다. (출처: Bloomberg)
    - 산업 전반의 생산 감축이나 확장 프로젝트 지연은 시장 재균형을 가속화할 것으로 예상됩니다. (출처: Discovery Alert)

출처: 1. https://www.bloomberg.com/news/articles/2024-11-14/lithium-in-nascent-recovery-but-longer-term-outlook-still-bleak 2. https://www.reuters.com/markets/commodities/lithium-prices-stabilise-2025-mine-closures-china-ev-sales-ease-glut-analysts-2025-01-13/ 3. https://www.reuters.com/markets/commodities/lithium-producer-albemarle-cut-workforce-lower-spending-2024-2024-01-17/ 4. https://www.fastmarkets.com/insights/lithium-weakness-pushes-chinese-producers-to-consider-output-cuts-sources/ 5. https://www.fastmarkets.com/insights/pilbara-minerals-lowers-production-2025/ 6. https://www.corelithium.com.au/assets/finniss-lithium-operation 7. http://lithium-chemical.com/Company/458.html

Greenbushes 리튬 광산 감산 및 운영 현황 (2023-2024)

카테고리: project_analysis

수집 시간: 2025-08-07 16:11:44

No topic

카테고리: No category

수집 시간: No time information

Greenbushes 리튬 광산 감산 및 운영 현황 (2023-2024)

기본 정보

- **프로젝트명**: Greenbushes
- **위치**: 서호주(Western Australia), 퍼스에서 남쪽으로 250km
- **운영사**: Talison Lithium (Tianqi Lithium Corporation 51%, Albemarle Corporation 49% 합작회사)
- **생산 규모**: 연간 약 1.95백만 톤의 리튬 스포듀민 생산 (세계 최대 경암 리튬 광산)
- **중요도**: 세계 리튬 스포듀민 수요의 약 1/3을 충족 가능

감산 관련 정보

2024년 감산 결정

- 2024년 1월, Greenbushes 광산은 리튬 시장 침체에 대응하여 생산량을 감축하기로 결정했습니다. (출처: ABC News, 2024년 1월 29일)
- 이는 전기차 수요 부진으로 인한 리튬 가격 하락에 따른 조치였습니다. (출처: Wikipedia)
- Talison의 대변인은 "Talison의 Greenbushes 리튬 사업은 40년 이상 리튬 산업을 선도해 왔으며 현재의 시장 변동에 대응할 수 있는 좋은 위치에 있다"고 언급했습니다. (출처: ABC News)

생산량 변화

- 2025 회계연도(FY2025)의 리튬 정광 생산 가이던스는 1.35-1.55백만 톤으로 설정되었습니다. (출처: Mysteel)
- 2025년 1분기 생산량은 342,000톤으로, 전분기 대비 13% 감소했으나 전년 동기 대비 22% 증가했습니다. (출처: Mysteel)
- 생산량 감소는 주로 낮은 원석 품위(feed grades)의 영향을 받았습니다. (출처: Mysteel)

판매 및 가격 동향

- 2025년 1분기 판매량은 366,000톤으로, 전분기 대비 17% 증가했으며 전년 동기 대비 두 배 증가했습니다. (출처: Mysteel)
- 일부 4분기 선적은 항만 혼잡으로 인해 1월로 지연되었습니다. (출처: Mysteel)
- 평균 FOB 판매 가격은 톤당 $791로 전분기 대비 7.47% 상승했습니다. (출처: Mysteel)
- 현금 비용은 톤당 AUD 341로 전분기 대비 5% 증가했으나 전년 동기 대비 12% 감소했습니다. (출처: Mysteel)

인력 관련 정보

- Talison의 대변인은 생산량 감축에도 불구하고 인력을 변경할 계획이 없다고 ABC에 밝혔습니다. (출처: ABC News, 2024년 1월 29일)

확장 계획

- 프로젝트는 일정에 따라 진행 중이며, 2025년 말까지 새로운 시설을 통해 첫 광석을 처리할 것으로 예상됩니다. (출처: Australian Mining, 2025년 7월 30일)

역사적 맥락

- Greenbushes의 리튬 광체는 1977년과 1980년 사이에 진행된 광범위한 시추 프로그램을 통해 확인되었으며, 광산은 30년 이상 리튬 광물을 생산하고 수출해 왔습니다. (출처: Talison Lithium 공식 웹사이트)
- 2018년 기준으로 Talison은 Greenbushes 광산을 총 용량의 60%로 운영하고 있었으며, 100% 용량으로 운영할 경우 전 세계 리튬 수요를 단독으로 충족시킬 수 있을 것으로 추정됩니다. (출처: Wikipedia)

출처: 1. https://en.wikipedia.org/wiki/Greenbushes_mine 2. https://www.abc.net.au/news/2024-01-29/greenbushes-to-reduce-production-in-lithium-downturn-igo/103402668 3. https://www.mysteel.net/analysis/5087659-overview-of-lithium-mining-projects-in-australia 4. https://www.talisonlithium.com/

Core Lithium의 Finniss 프로젝트 폐쇄 및 재개 계획 (2023-2024)

카테고리: project_analysis

수집 시간: 2025-08-07 16:12:41

No topic

카테고리: No category

수집 시간: No time information

Core Lithium의 Finniss 프로젝트 폐쇄 및 재개 계획 (2023-2024)

기본 정보

- **프로젝트명**: Finniss Lithium Operation
- **위치**: 호주 노던 테리토리(Northern Territory), 다윈 항구에서 남서쪽으로 88km
- **운영사**: Core Lithium Ltd (100% 소유)
- **프로젝트 규모**: 500km² 이상의 광구 면적
- **광산 유형**: 노천 및 지하 광산

폐쇄 관련 정보

단계적 폐쇄 과정

1. **2023년 말**: 리튬 가격 하락과 비용 절감에 초점을 맞추면서 BP33(Finniss 지역 내 지하 프로젝트)의 초기 작업을 중단했습니다. 우기 채굴 조건으로 인한 복잡성도 중단 결정에 영향을 미쳤습니다. (출처: ChemAnalyst)
2. **2024년 1월**: 리튬 가격 하락으로 인해 Grants 노천 광산의 채굴을 중단했습니다. (출처: Core Lithium 공식 웹사이트, Mining Technology)
3. **2024년 중반**: 가공 시설 운영을 중단하고 모든 인프라를 관리 상태(care and maintenance)로 전환했습니다. (출처: Core Lithium 공식 웹사이트)

폐쇄 사유

- 리튬 가격의 급격한 하락
- 글로벌 리튬 공급 과잉
- 운영 비용 증가
- 우기 채굴 조건으로 인한 어려움 (출처: ChemAnalyst, Mining Technology)

재개 계획

재개 연구(Restart Study)

- Core Lithium은 Finniss 리튬 사업의 재개 연구가 계획대로 진행 중이며, 2025년 6월 분기에 완료될 것으로 발표했습니다. (출처: Mining Technology, ChemAnalyst)

재개 전략

- 재개 연구는 기존 인프라를 최적화하고 향후 채굴 및 가공 활동을 단순화하여 현장의 생산성을 향상시키는 것을 목표로 합니다. (출처: Mining Technology)
- Core는 더 낮은 운영 비용, 향상된 효율성 및 더 탄력적인 비즈니스 구조를 만들기 위해 전략적 움직임을 취하고 있습니다. (출처: Mining Technology)

계약 및 자산 관리

- Core는 이전 운영 계약업체로부터 "있는 그대로, 있는 곳에서(as is, where is)" 기준으로 자산을 인수할 예정입니다. (출처: Mining Technology)
- 총 A$19.5백만에 달하는 이러한 계약 문제의 해결은 2025년 7월까지 완료될 예정이며, Core의 기존 현금 준비금으로 충당될 것입니다. (출처: Mining Technology)

오프테이크 계약 관리

- Core Lithium은 재개 경로를 명확히 하기 위해 중국 Yahua와의 오프테이크 계약을 종료했습니다. 이는 회사가 재개 계획을 더 유연하게 추진할 수 있게 해줍니다. (출처: ChemAnalyst)

투자 및 개발 현황

- Core Lithium은 Finniss 프로젝트의 Grants 및 BP33 리튬 매장지에 A$2억 5천만(US$1억 5,760만) 이상을 투자했습니다. (출처: Mining Technology)
- 2021년 9월, Core Lithium의 이사회는 노던 테리토리에 위치한 완전 소유 Finniss 프로젝트 개발을 시작하기로 최종 투자 결정을 내렸습니다. (출처: Tesmanian)

미래 전망

- 2025년 재개 연구는 Finniss가 장기적이고 저비용의 프로젝트이며 생산 능력이 증가했음을 확인했습니다. (출처: Core Lithium 투자자 개요)
- Core Lithium은 노던 테리토리와 남호주 전역에 탐사 지역도 보유하고 있어 향후 성장 잠재력을 갖추고 있습니다. (출처: Core Lithium 투자자 개요)

출처: 1. https://www.mining-technology.com/news/core-lithium-finniss-restart-study/ 2. https://www.corelithium.com.au/ass lithium-operation 3. https://www.chemanalyst.com/NewsAndDeals/NewsDetails/core-lithium-clears-path-for-finniss-restart-with-yahua-offtake-exit-36544

Pilbara Minerals의 Pilgangoora 프로젝트 감산 및 Ngungaju 공장 폐쇄 (2024)

카테고리: project_analysis

수집 시간: 2025-08-07 16:13:34

No topic

카테고리: No category

수집 시간: No time information

Pilbara Minerals의 Pilgangoora 프로젝트 감산 및 Ngungaju 공장 폐쇄 (2024)

기본 정보

- **프로젝트명**: Pilgangoora

- **위치**: 서호주(Western Australia)
- **운영사**: Pilbara Minerals
- **중요도**: 호주 최대 순수 리튬 광산 기업

Ngungaju 공장 폐쇄 결정

폐쇄 시점 및 내용

- **발표 시점**: 2024년 10월 29일, Pilbara Minerals는 Ngungaju 공장을 관리 상태(care and maintenance)로 전환한다고 발표했습니다. (출처: Fastmarkets, 2024년 10월 30일)
- **폐쇄 시점**: 2024년 12월부터 Ngungaju 공장 운영 중단 (출처: Argus Media)
- **폐쇄 기간**: 2025-26 회계연도 전체 기간 동안 관리 상태 유지 계획 (출처: Argus Media)

폐쇄 사유

- 리튬 가격 약세로 인한 조치 (출처: Argus Media)
- 시장 공급 과잉 상황에서 생산 감축 필요성 (출처: Fastmarkets)
- "시장이 공급 과잉 상태이므로 생산 감축이 필요하며, Pilbara의 Ngungaju 공장 가동 중단 결정은 중요한 발전입니다." - William Adams, Fastmarkets 베이스 메탈 및 배터리 원자재 연구 책임자 (출처: Fastmarkets)

영향 및 조치

- **인력 조정**: 약 50명의 직원 감축과 나머지 50명은 다른 운영 부문으로 재배치 (출처: Australian Financial Review)
- **재정적 영향**: 이 조치로 약 A$2억(약 1억 3,120만 달러)의 현금 흐름 개선 예상 (출처: Fastmarkets)
- **생산 능력 영향**: Ngungaju 공장은 연간 180,000-200,000톤의 스포듀민 생산 능력 보유 (출처: Argus Media)

전략적 결정 배경

- "Ngungaju 공장은 Altura Mining으로부터 인수한 이후 상당한 업그레이드를 거쳤지만, Pilgan 공장의 규모나 가공 능력에 미치지 못합니다." - Pilbara Minerals 성명 (출처: Fastmarkets)
- 회사는 Pilgan 공장에만 집중하기로 결정 (출처: Fastmarkets)

전체 생산 현황 및 전망

2024-25 회계연도 생산 실적

- **총 생산량**: 754,600톤의 스포듀민 생산 (전년 대비 4% 증가) (출처: Argus Media)
- **가이던스 초과**: 700,000-740,000톤 가이던스 범위의 상한선 초과 (출처: Argus Media)
- **운영 비용**: A$627/톤(FOB 기준)으로 2023-24년의 A$654/톤에서 감소 (출처: Argus Media)

2025-26 회계연도 전망

- **생산 계획**: 820,000-870,000톤의 스포듀민 생산 계획 (출처: Argus Media)
- **비용 전망**: A$560/톤 - A$600/톤(FOB 기준)의 단위 운영 비용 예상 (출처: Argus Media)

확장 프로젝트 및 최적화

- **P1000 확장**: 2024년 1-3월에 P1000 확장 프로젝트 시운전, 4-6월에 증설 및 최적화 완료 (출처: Argus Media)
- **P850 운영 모델**: 2024년 9월에 P850 운영 모델로 전환하여 추가 비용 절감 달성 (출처: Australian Resources and Investment, 2025년 4월 22일)
- **장기 목표**: 연간 100만 톤의 스포듀민 생산을 목표로 함 (출처: Australian Resources and Investment)

시장 대응 전략

비용 절감 및 보존 조치

- 2023년 9월 이후 배당금 지급 중단 (출처: Australian Resources and Investment)
- 2024년 3월 인력 감축 시행 (출처: Australian Resources and Investment)
- 2024년 9월 P850 운영 모델로 전환하여 추가 비용 절감 (출처: Australian Resources and Investment)

성장 이니셔티브 유지

- 리튬 가격 약세에도 불구하고 성장 이니셔티브 우선순위 유지 (출처: Australian Resources and Investment)
- 확장된 Pilgan 공장을 통해 2025-26 회계연도에 더 높은 생산량과 낮은 단위 비용 달성 계획 (출처: Australian Resources and Investment)

시장 변동성 대응

- "일관되고 규율 있는" 전략을 통해 지속적인 시장 변동성에 대응 (출처: Australian Resources and Investment)
- 열대성 사이클론 Zelia로 인한 6일간의 생산 손실에도 불구하고 운영 유지 (출처: Australian Resources and Investment)

출처: 1. https://www.fastmarkets.com/insights/pilbara-minerals-lowers-production-2025/ 2. https://www.argusmedia.com/en and-insights/latest-market-news/2715512-pilbara-minerals-lifts-spodumene-output-by-4pc-in-fy25 3. https://www.afr.com/c minerals-cuts-lithium-output-suspends-plant-20241030-p5kmgz 4. https://www.australianresourcesandinvestment.com.au/2 pilbara-minerals-is-navigating-the-lithium-downturn/

Albemarle의 생산 감축 및 인력 조정 (2023-2024)

카테고리: project_analysis

수집 시간: 2025-08-07 16:15:09

No topic

카테고리: No category

수집 시간: No time information

Albemarle의 생산 감축 및 인력 조정 (2023-2024)

기본 정보

- **기업명**: Albemarle Corporation
- **위치**: 미국 기반, 글로벌 운영 (Silver Peak - 미국 네바다, Salar de Atacama - 칠레 등)
- **중요도**: 세계 최대 리튬 생산업체

비용 절감 및 인력 조정 조치

2024년 1월 발표된 주요 조치

- **인력 감축**: 2024년 1월 29일 보도에 따르면, Albemarle는 전체 인력의 약 4%를 감축했습니다. (출처: Reuters, 2024년 1월 29일)
- **비용 절감 목표**: 이러한 조치로 2024년에만 최소 5,000만 달러의 비용 절감을 예상했습니다. (출처: Reuters)
- **지출 연기**: 미국 리튬 정제소 프로젝트에 대한 지출을 연기하기로 결정했습니다. (출처: Reuters, 2024년 1월 17일)

추가 비용 절감 전략

- **자본 지출 감소**: 2024년 자본 지출 전망을 전년 대비 3억-5억 달러 감소시켰습니다. (출처: Albemarle 투자자 프레젠테이션, 2024년 5월)
- **운영 비용 절감**: 연간 운영 비용을 약 9,500만 달러 감소시키는 계획을 시행, 주로 SG&A(판매, 일반 및 관리 비용) 부문에서 절감했습니다. (출처: Albemarle 투자자 프레젠테이션)
- **계약 서비스 지출 감소**: 인력 감축과 함께 계약 서비스에 대한 지출도 줄였습니다. (출처: Albemarle 투자자 프레젠테이션)

프로젝트 재조정

- **대규모 프로젝트 재단계화**: 상당히 진행되었거나, 완료에 가깝거나, 시작 단계에 있는 프로젝트에 집중하기 위해 대규모 프로젝트를 재조정했습니다. (출처: Albemarle 투자자 프레젠테이션)
- **비핵심 투자 매각 검토**: 비핵심 투자 매각을 평가하고 있으며, Liontown 지분을 현금화했습니다. (출처: Albemarle 투자자 프레젠테이션)

주요 프로젝트 현황

Silver Peak (미국 네바다)

- **중요성**: 현재 미국 내 리튬 생산의 거의 전부를 담당하는 유일한 리튬 생산 시설입니다. 미국 리튬 생산은 세계 생산량의 2% 미만을 차지합니다. (출처: Albemarle 공식 웹사이트)
- **위치**: 라스베이거스에서 북서쪽으로 약 200마일 떨어진 곳에 위치합니다. (출처: Albemarle 공식 웹사이트)
- **생산 방식**: "완벽한 증발 환경"을 활용하여 리튬이 풍부한 염수에 태양 증발 공정을 사용합니다. (출처: Albemarle 공식 웹사이트)
- **생산 감축 관련 정보**: 구체적인 생산 감축 정보는 공개되지 않았으나, 전반적인 비용 절감 조치의 일환으로 영향을 받았을 것으로 추정됩니다.

Salar de Atacama (칠레)

- **중요성**: Albemarle의 핵심 자산으로, 세계에서 가장 효율적인 리튬 염수 매장지 중 하나에 50% 지분을 보유하고 있습니다. (출처: AInvest)
- **비용 경쟁력**: 중국 경쟁사들이 수익성을 위해 톤당 15,000달러 이상의 리튬 가격을 필요로 하는 반면, Albemarle의 칠레 운영은 톤당 5,000달러 미만의 현금 비용 구조를 유지하고 있어 70%의 비용 우위를 가지고 있습니다. (출처: AInvest)
- **계약 기간**: Albemarle의 Salar de Atacama 계약은 2043년에 만료됩니다. (출처: Albemarle 공식 웹사이트)
- **재정 기여**: 2022년에 칠레 정부에 6억 5천만 달러를 기여했으며, 2023년에는 그 이상을 기여할 것으로 발표했습니다. (출처: Albemarle 공식 웹사이트)

시장 상황 및 전망

리튬 시장 상황

- **가격 하락**: 2023년 초부터 리튬 가격이 74% 하락하여 톤당 18,500달러 수준에 이르렀습니다. (출처: AInvest)
- **수요 전망 조정**: 2030년 리튬 수요 전망이 이전 예측보다 약 10% 하락했으며, 이는 OEM 발표, 배터리 크기 성장 완화, 다운스트림 리튬 염 재고 감소 등의 요인 때문입니다. (출처: Albemarle 투자자 프레젠테이션)

공급 전망

- **공급 균형**: 최근 발표된 생산 감축, 시작 지연, 확장 지연 등으로 인해 공급이 균형을 유지할 것으로 예상됩니다. (출처: Albemarle 투자자 프레젠테이션)
- **수요 충족을 위한 성장 인센티브**: 수요를 충족시키기 위한 성장을 장려하려면 더 높은 가격이 필요합니다. (출처: Albemarle 투자자 프레젠테이션)

효율성 개선 이니셔티브

2024년 목표

- **제조 개선**: La Negra, Qinzhou, Xinyu에서의 수율, 속도 및 가용성 개선을 통해 8,000만 달러 절감 목표 (출처: Albemarle 투자자 프레젠테이션)
- **조달 최적화**: 주요 원자재 및 물류, 특히 해상 및 도로 운송에 대한 비용 절감을 통해 1억 5천만 달러 절감 목표 (출처: Albemarle 투자자 프레젠테이션)
- **백오피스 효율화**: 구조조정 비용 절감, 제3자 지출 감소를 통해 5,000만 달러 절감 목표 (출처: Albemarle 투자자 프레젠테이션)

현금 관리

- **운전자본 관리**: 재고 관리, 광산에서 고객까지의 공급망 압축 (출처: Albemarle 투자자 프레젠테이션)
- **2024년 목표 초과**: 2024년 1분기 기준으로 2억 8천만 달러의 목표를 초과할 것으로 예상됩니다. (출처: Albemarle 투자자 프레젠테이션)

미래 기술 투자

직접 리튬 추출(DLE) 기술

- **상용화 전망**: DLE 기술의 상용화는 3-5년 후로 예상되며, 이는 Albemarle이 다음 리튬 사이클을 주도할 수 있는 위치에 있게 합니다. (출처: AInvest)
- **비용 절감 잠재력**: DLE가 2027년까지 칠레 생산 비용을 톤당 3,000달러로 낮출 경우, 경쟁사들이 톤당 10,000달러 이상의 손익분기점에서 어려움을 겪는 동안 Albemarle의 마진이 확대될 것입니다. (출처: AInvest)

출처: 1. https://www.reuters.com/markets/commodities/lithium-producer-albemarle-cut-workforce-lower-spending-2024-2024-01-17/ 2. https://www.reuters.com/markets/commodities/albemarle-lays-off-4-its-total-workforce-information-2024-01-29/ 3. https://s201.q4cdn.com/960975307/files/doc_presentations/2024/May/01/2024_05-alb-investor-presentation-vweb.pdf 4. https://www.ainvest.com/news/albemarle-strategic-resilience-navigating-lithium-downturn-focus-efficiency-innovation-2506/ 5. https://www.albemarle.com/us/en/news/silver-peak-stands-alone-us-lithium-resource

SQM의 Salar de Atacama 리튬 사업 감산 및 인력 감축 (2023-2025)

카테고리: project_analysis

수집 시간: 2025-08-07 16:16:26

No topic

카테고리: No category

수집 시간: No time information

SQM의 Salar de Atacama 리튬 사업 감산 및 인력 감축 (2023-2025)

기본 정보

- **기업명**: SQM (Sociedad Química y Minera de Chile S.A.)
- **프로젝트**: Salar de Atacama 리튬 사업
- **위치**: 칠레 아타카마 사막
- **중요도**: 세계 2위 리튬 생산업체

인력 감축 및 비용 절감 조치

인력 감축

- **2025년 6월** SQM은 장기화된 리튬 가격 하락에 대응하기 위해 칠레 인력의 5%를 감축하기 시작했습니다. (출처: Reuters, 2025년 6월 25일)
- 이는 리튬 가격 하락과 시장 공급 과잉 상황에 대응하기 위한 조치였습니다.

재정 상황

- **2024년 1분기** SQM은 공급 과잉으로 인한 가격 압박과 11억 달러의 세금 판결 영향으로 8억 7천만 달러의 순손실을 기록했습니다. (출처: Reuters, 2024년 5월 23일)
- **2024년 연간 실적** 리튬 및 파생상품의 총 매출은 22억 4,130만 달러로, 2023년의 52억 달러에서 56.7% 급감했습니다. (출처: SQM 2024년 4분기 실적 발표)
- **2024년 관리비용** 총 1억 8,600만 달러(매출의 4.1%)로 집계되었습니다. (출처: SQM 2024년 4분기 실적 발표)

생산 전략 변화

- SQM은 2024년 1분기에 판매량을 27% 증가시켜 가격 하락을 상쇄하려 했으나, 경영진은 2분기에 더 낮은 가격을 예상했습니다. (출처: Americas Quarterly)
- 이는 가격 하락 상황에서 물량을 늘려 수익을 유지하려는 전략이었습니다.

사업 확장 및 미래 계획

Codelco와의 파트너십

- SQM은 지정학적 혼란 시기에 시장 우위를 확보하기 위해 칠레 국영 구리 광산 기업인 Codelco와의 장기 파트너십에 미래를 걸고 있습니다. (출처: Americas Quarterly)
- 이 예비 공공-민간 협력은 2024년 5월에 공개 입찰이 아닌 직접 협상을 통해 체결되었으며, 2024년 11월 칠레 선거를 앞두고 점점 더 많은 조사를 받고 있습니다. (출처: Americas Quarterly)

생산 확대 계획

- 칠레에서 SQM은 수산화리튬 생산을 현재 35,000 메트릭 톤(MT)에서 2024년 말까지 100,000 MT로, 탄산리튬을 185,000 MT에서 2026년까지 240,000 MT로 확대할 계획입니다. (출처: Americas Quarterly)
- 2023년 기준으로 리튬 탄산염 생산 능력은 180,000 메트릭 톤이었습니다. (출처: SQM Litio 공식 웹사이트)

Salar Futuro 프로젝트

- SQM과 Codelco 파트너십은 "염호 주변 지역사회와의 지속적인 관계를 촉진하기 위한 거버넌스 모델 구축"과 "더 효율적이고 지속 가능한 생산, 즉 더 적은 염수와 대륙 물을 사용하지 않고 더 많은 리튬을 생산하는" 새로운 추출 방법 구현을 약속하는 "Salar Futuro" 프로젝트를 추진하고 있습니다. (출처: Grist)

계약 연장

- SQM은 Codelco와 협약을 체결하여 2060년까지 Salar de Atacama에서의 운영을 계속할 수 있게 되었습니다. 두 회사는 운영을 위한 새로운 법인을 설립할 예정이며, Codelco가 회사의 50% 플러스 1주를 소유하게 됩니다. (출처: Investing News Network)

지속가능성 및 환경 영향

물 사용 감소 약속

- SQM은 아타카마 염호에서의 염수 추출을 2030년까지 50% 줄이기로 약속했으며, 이는 안토파가스타의 리튬 정제소 확장 계획에 영향을 미치지 않을 것이라고 밝혔습니다. (출처: SQM Litio 지속가능한 리튬 생산 보고서)
- 이러한 감축은 운영을 최적화함으로써 달성될 예정입니다.

환경 친화적 추출 공정

- SQM의 Salar de Atacama에서의 리튬 추출 공정은 환경 친화적이며 리튬의 지속 가능한 개발 및 생산에 중점을 두고 있습니다. (출처: SQM Litio 공식 웹사이트)

IRMA 감사

- 2025년에는 SQM Salar의 Salar de Atacama 리튬 운영에 대한 IRMA(Initiative for Responsible Mining Assurance) 감사가 예정되어 있습니다. (출처: SQM Litio 공식 웹사이트)

정치적 맥락 및 국가 전략

칠레의 리튬 국가 전략

- 가브리엘 보리치 대통령은 Codelco-SQM 제휴가 칠레의 리튬 생산을 가속화하고, 정부 수입을 증가시키며, 귀중한 자원에 대한 국가 통제를 부과하고, 기술적 노하우를 이전하며, 물 사용에 대한 우려가 커짐에 따라 지속 가능성을 개선할 것이라고 기대하고 있습니다. (출처: Americas Quarterly)

비판적 시각

- 비판론자들은 Codelco-SQM 거래가 투명성이 부족하며, 칠레가 경쟁 입찰과 리튬 채굴을 더 경험이 많고 효율적인 민간 부문에 맡기는 양허를 승인함으로써 더 많은 임대료를 확보할 수 있다고 주장합니다. (출처: Americas Quarterly)
- "SQM이 칠레에서 더 성장하는 것은 사실상 불가능했다"고 산티아고 기반의 리튬 컨설턴트 다니엘 히메네스는 말했습니다. (출처: Americas Quarterly)

출처: 1. https://www.reuters.com/world/americas/chilean-lithium-miner-sqm-begins-laying-off-5-local-workforce-2025-06-25/ 2. https://www.reuters.com/markets/commodities/lithium-miner-sqm-swings-870-million-loss-sinking-prices-tax-ruling-2024-05-23/ 3. https://ir.sqm.com/static-files/12d462b5-5953-4cfd-b7c4-ae8a571e739c 4. https://www.americasquarterly.org/article/chiles-embattled-lithium-king/ 5. https://sqmlitio.com/en/

No topic

카테고리: No category

수집 시간: No time information

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카테고리: No category

수집 시간: No time information

Lithium Project Production Cuts and Closures in Australia (2023-2024)

카테고리: production_cuts_closures

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Lithium Project Production Cuts and Closures in Australia (2023-2024)

Core Lithium - Finniss Project

- **Status**: Operations suspended in January 2024
- **Details**: Core Lithium halted mining operations at Finniss on January 5, 2024, due to sustained lithium price declines
- **Background**: The Mt Finniss project was designed for 197,000 tonnes of annual spodumene production with an eight-year mine life
- **Current Activity**: Final sales expected in Q3 2024, with residual inventory anticipated to be cleared by June
- **Cost Impact**: While full-process cash costs approached $1,000/tonne, Q2 2024 costs fell to $420/tonne with only stockpile processing underway
- **Historical Context**: Core Lithium secured initial tenure over the Bynoe pegmatite field in 2016 and released its first mineral resource estimate in mid-2017. Within six years, the project reached a final investment decision and secured long-term offtake agreements with Sichuan Yahua and Ganfeng Lithium.

Pilbara Minerals - Ngungaju Plant

- **Status**: Placed into care and maintenance in 2024
- **Details**: Pilbara Minerals suspended one of its two processing plants in Western Australia and reduced production
- **Workforce Impact**: Approximately 50 workers were shed as part of the decision to mothball the plant, with the remaining 50 employees redeployed across operations
- **Background**: Pilgangoora began production in 2018, with Ngungaju (purchased from bankruptcy and restarted in 2021) placed into care and maintenance in 2024 due to low prices
- **Production Capacity**: Pilgangoora had 1,000,000 tonnes per annum capacity, while Ngungaju had 200,000 tonnes capacity

Talison Lithium - Greenbushes Mine

- **Status**: Production cuts announced
- **Details**: Cut of approximately 100,000 tonnes
- **Background**: Greenbushes Mine had 1.5 million tonnes per annum capacity and was undergoing expansion to 2 million tonnes per annum before the cuts
- **Ownership**: Owned by Albemarle

Liontown Resources - Kathleen Valley

- **Status**: Deferred expansion
- **Details**: Liontown Resources announced it could defer the expansion and ramp-up of its Kathleen Valley lithium project as part of a cost-cutting review
- **Background**: Kathleen Valley began production in 2024 with planned capacity of 500,000 tonnes per annum
- **Context**: Spot price slump led to revisiting the ramp-up strategy

Overall Australian Production Impact

- Australian production in Q1 2024 declined by about 12.3% compared to the previous quarter
- Production cuts in Australia started in January 2024 but built momentum throughout the year
- Several miners announced production cuts, plans to place plants on care and maintenance, and suspension of planned expansions due to market conditions
- The reduction in production was mainly due to constraints on high-cost production capacity

출처: 1. http://lithium-chemical.com/Company/458.html 2. https://www.mysteel.net/analysis/5087659-overview-of-lithium-mining-projects-in-australia 3. https://www.fastmarkets.com/insights/facing-the-tightening-lithium-supply-challenge-in-2025/ 4. https://www.sciencedirect.com/science/article/pii/S2214790X25000814 5. https://www.afr.com/comp minerals-cuts-lithium-output-suspends-plant-20241030-p5kmgz

Lithium Project Production Cuts and Closures in China (2023-2024)

카테고리: production_cuts_closures

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수집 시간: No time information

Lithium Project Production Cuts and Closures in China (2023-2024)

Major Lepidolite Mine Closure

- **Status**: Production halted in September 2024
- **Details**: A major Chinese producer halted output at one of the biggest lepidolite mines in the country
- **Impact**: Caused three associated lepidolite processing plants to shut down
- **Context**: This was one of the first significant production cuts in China, occurring later than cuts in Western countries

Chinese Production Strategy During Price Crash

- **Response to 2023 Price Crash**: While non-Chinese companies pulled back production volume significantly, Chinese-owned companies pulled back much less and in some cases continued to expand production
- **Result**: Generated an oversupply of lithium, cobalt, nickel, and natural and synthetic graphite
- **Government Support**: Chinese government policies enabled these decisions by making it financially viable for Chinese companies to sell materials at low prices that would otherwise be unsustainable

Chinese Lithium Production Growth

- Despite market challenges, China's lithium supply grew by nearly 15% year-on-year, from 35,700 metric tons in 2023 to 41,000 metric tons in 2024
- China maintained its position as the third-largest lithium producer globally in 2024

Inventory Levels

- By the end of 2024, Chinese lithium salt inventories stood at approximately 220,000 tonnes of LCE (Lithium Carbonate Equivalent)
- This represented about 20% of annual demand, indicating significant oversupply
- Post-Chinese Lunar New Year holiday restocking failed to meet expectations due to slow downstream buying activity and high upstream inventory levels

Market Dynamics

- Chinese producers were not immune to weakening prices in 2024

- Higher-cost Chinese operations, particularly lepidolite mines with higher production costs, were the first to implement cuts
- The market showed relief when the major lepidolite mine halted production in September 2024
- Despite some cuts, China's overall strategy appeared to focus on maintaining market share rather than balancing supply with demand

출처: 1. https://www.fastmarkets.com/insights/facing-the-tightening-lithium-supply-challenge-in-2025/ 2. https://fas.org/pub thinking-on-critical-minerals/ 3. https://investingnews.com/daily/resource-investing/battery-metals-investing/lithium-investing/lithium-market-update/

Lithium Project Developments in South America (2023-2024)

카테고리: production_expansions

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Lithium Project Developments in South America (2023-2024)

Chile

SQM (Sociedad Química y Minera de Chile)

- **Current Operations**: Operates in Salar de Atacama
- **Production Capacity**: Produces both lithium carbonate and lithium hydroxide
- **Expansion Plans**:
  - Plans to expand production of lithium hydroxide from 35,000 metric tons to 100,000 metric tons by the end of 2024
  - Plans to increase lithium carbonate production from 185,000 metric tons to 240,000 metric tons by 2026
- **License Status**: Has license to extract lithium until 2030
- **Financial Performance**: Reported revenues of $7.5 billion in 2023 and $4.5 billion in 2024
- **Government Relations**: In negotiations with Codelco (state-owned copper mining company) to extend the current contract beyond 2030
  - A preliminary public-private deal was signed in May 2024 after direct negotiations
  - A new company would be created between SQM and Codelco, with Codelco controlling it, to expand lithium production and improve yields per volume of brine extracted

Albemarle

- **Current Operations**: Operates in Atacama and La Negra
- **Activities**: Develops conventional mining and activities for refining (technical-grade and battery-grade lithium carbonate)
- **Additional Facilities**: Maintains R&D laboratories
- **License Status**: Has license to extract lithium until 2043
- **Government Relations**: No indications that the contract between the Chilean state and Albemarle will be affected by the new lithium policy

Chilean Government Lithium Policy (April 2023)

- **Nationalization Plans**: The Chilean government announced plans for nationalizing its lithium industry in April 2023
- **Impact**: The announcement primarily affected SQM and Albemarle
- **Market Response**: Shares of SQM in the Santiago Stock Exchange dropped by 15% following the announcement
- **Policy Objectives**:
  1. Increasing production in Chile's only two active lithium operations (SQM and Albemarle) in the Atacama Salt Flat
  2. Inviting new players to invest in other salt flats outside Atacama

- **Government Control**: The main goal was to control lithium operations, either directly or through state-owned enterprises like Codelco and Enami
- **New Developments**: In 2024, the government invited foreign investors to exploit other salt flats, with nearly 50 companies responding

Argentina

Historical Context

- **First Operations**: Lithium extraction began in 1997 when the U.S.-based FMC Corporation initiated operations at the Salar del Hombre Muerto in Catamarca province
- **Recent Expansion**: In the 21st century, new projects were launched in Jujuy, Salta, and Catamarca provinces, particularly at the Cauchari-Olaroz and Pastos Grandes salt flats

Current Status

- **Global Position**: Argentina is the world's fifth-largest lithium producer
- **Production Growth**: Achieved an 87.5% increase in annual production in 2024
- **Production Volume**: In 2022, Argentina produced 33,000 tonnes of LCE (Lithium Carbonate Equivalent), making it the second-largest producer in the region and fourth globally

Overall South American Production

- **Chile's Production Growth**: Chile expanded output at its existing operations by 80% since 2016 to about 140,000 tons annually (as of 2022)
- **Regional Dominance**: All lithium extracted in Chile as of 2023 comes from Salar de Atacama
- **Bolivia's Status**: Bolivia's production was limited, reaching only 600 tonnes of LCE in 2023
- **Production Methods**: Traditional evaporation methods are being used, though Direct Lithium Extraction (DLE) technology is beginning to be adopted to reduce production times and environmental impacts

출처: 1. https://www.sciencedirect.com/science/article/pii/S2214790X24001771 2. https://en.wikipedia.org/wiki/Lithium_Tria 3. https://www.americasquarterly.org/article/chiles-embattled-lithium-king/ 4. https://www.energypolicy.columbia.edu/chiles state-centric-lithium-policy-may-deter-investment/ 5. https://iwgia.org/en/news/5777-a-just-energy-transition-the-impacts-of-lithium-extraction-on-the-andean-salt-flats-of-argentina,-bolivia,-and-chile.html

Global Lithium Subsidies and Government Support (2023-2024)

카테고리: subsidies_government_support

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Global Lithium Subsidies and Government Support (2023-2024)

United States

Inflation Reduction Act (IRA)

- **Critical Mineral Requirements**: To be eligible for a $3,750 credit, a specific percentage of the value of critical minerals (including lithium) contained in EV batteries must be extracted or processed in the United States or a country with which the United States has a free trade agreement, or be recycled in North America
- **Percentage Requirements by Year**:
  - 2023: 50%
  - 2024 and 2025: 60%
  - 2026: 70%
  - 2027: 80%

    - 2028: 90%
    - From 2029: 100%
- **Investment Impact**: The IRA provided subsidies for battery production and clean energy initiatives, generating over $224 billion in investments
- **Project Support**: In August 2019, a lithium project was initiated, but it was not until 2024 that the Department of Energy finalized a loan to support its development, including the construction of a sulfuric acid plant and a lithium processing facility

Processing Capacity Prioritization

- **Focus Areas**: The U.S. government has prioritized the expansion of processing capacity for lithium, graphite, cobalt, and nickel
- **Market Dynamics**: Demand from domestic battery manufacturing is expected to outpace the current planned capacity for all these materials
- **Support Distribution**: Tariffs and tax incentives have resulted in favorable pricing for lithium and graphite project developers, but cobalt and nickel processing has received less support and attention

Project Examples

- **Controlled Thermal Resources**: In January 2024, began construction of their Salton Sea lithium mine and geothermal power plant that will initially produce 25,000 metric tons of lithium, with potential for up to 175,000 metric tons once completed
- **Kings Mountain Processing Facility**: Albemarle's facility in North Carolina processes domestically extracted lithium, creating a closed domestic supply loop

China

Government Support for Overproduction

- **Production Strategy**: After the price crash in 2023, while other companies pulled back production significantly, Chinese-owned companies pulled back much less and in some cases continued to expand production
- **Government Role**: Chinese government policies enabled these decisions by making it financially viable for Chinese companies to sell materials at low prices that would otherwise be unsustainable
- **Market Impact**: This strategy generated an oversupply of lithium, cobalt, nickel, and natural and synthetic graphite
- **Global Reach**: Chinese companies are the primary financial backers for 80% of both the Democratic Republic of the Congo's cobalt mines and Indonesia's nickel mines
- **Lithium Acquisition**: Chinese companies have bought half of all the lithium mines offered for sale since 2018, in addition to domestically mining 18% of global lithium

EV Subsidies

- **Subsidy Expiration**: The Chinese Government's decade-long program of subsidies for electric vehicle (EV) purchases expired in 2023
- **Market Impact**: This expiration, combined with concern of a short-term lithium oversupply and weaker-than-expected EV sales, contributed to price volatility

Global Policy Trends

Government Initiatives

- **Clean Energy Transition**: Government initiatives aimed at accelerating the clean energy transition are playing a significant role in boosting lithium demand
- **Policy Examples**: Policies like the Inflation Reduction Act (IRA) in the U.S. and the European Critical Raw Materials Act offer financial incentives, subsidies, and tax breaks for:
    - EV production and purchases
    - Investments in raw material sourcing
    - Battery manufacturing
    - Renewable energy infrastructure

Australia's Critical Minerals Strategy

- **Policy Framework**: The Australian Government's Critical Minerals Strategy (2023) emphasizes the importance of developing domestic resources to support both national interests and global clean energy transitions
- **Project Support**: By bringing additional lithium resources into production, projects like Core Lithium's Finniss support broader policy goals of positioning Australia as a reliable, ethical supplier of materials essential to decarbonization
- **Environmental Focus**: The strategy notes that "Critical minerals projects that demonstrate superior environmental and social practices are increasingly favored by both regulators and investors"

출처: 1. https://fas.org/publication/critical-thinking-on-critical-minerals/ 2. https://home.treasury.gov/news/press-releases/jy1939 3. https://warontherocks.com/2025/04/a-federal-critical-mineral-processing-initiative-securing-u-s-mineral-independence-from-china/ 4. https://lithiumharvest.com/knowledge/lithium/the-lithium-mining-market/ 5. https://councilonstrategicrisks.org/2025/05/30/the-devil-is-in-the-details-minerals-batteries-and-us-dependence-on-chinese-imports/

Global Lithium Market Trends and Price Dynamics (2015-2024)

카테고리: market_trends

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Global Lithium Market Trends and Price Dynamics (2015-2024)

Price Volatility and Crash (2022-2024)

- **Peak Prices**: Spot prices reached over $80,000 per ton in December 2022
- **Price Collapse**: By January 30, 2024, prices had fallen to just over $13,000 per ton, a decline of over 80%
- **Causes of Price Crash**:
  - Oversupply in the market
  - Softening demand, particularly from the EV sector
  - Slower than expected rates of electric vehicle adoption
  - Decreased EV production targets in Europe and the US

Supply and Demand Dynamics

- **Production Growth**: Global lithium production jumped from just over 737,000 tonnes in 2022 to almost 1.2 million tonnes in 2024 on a lithium carbonate equivalent (LCE) basis
- **Market Surplus**: Fastmarkets' research team estimated that the lithium market recorded a surplus of around 175,000 tonnes in 2023, and almost 154,000 tonnes in 2024
- **EV Adoption Slowdown**: The EV penetration rate in the US averaged 6.8% in the first 6 months of 2024, compared to 7.5% in 2023
- **Market Forecast**: The global lithium-ion battery market is expected to stay oversupplied until 2028 due to a decline in EV production targets in the US and EU

Production Cuts and Project Impacts

- **Timing of Cuts**: Significant production cuts did not emerge until September 2024, when a major Chinese producer halted output at one of the biggest lepidolite mines
- **Cascading Effects**: This closure caused three associated lepidolite processing plants to shut down
- **Western Response**: Further production cut announcements emerged in the last quarter of 2024, along with project start-up delays and curtailments, mainly in Western countries
- **Australian Impact**: Higher-cost Australian projects chose to push projects into care and maintenance rather than operate at a loss
- **Specific Cuts in 2024**:

- Finiss (Core Lithium): Operations suspended, processing stockpiles from concentrator (280kt)
- Talison (Greenbushes): Cut of approximately 100kt
- Kathleen Valley (Liontown): Spot price slump led to revisiting ramp-up strategy

Price Stabilization and Future Outlook

- **Price Convergence**: Lithium prices in China and North America have converged, with North American lithium no longer enjoying the premium it did in 2023 and the first half of 2024
- **Medium-Term Forecast**: Analysts expect lithium carbonate to stay below $20 per kg globally at least through 2027
- **S&P Global Projection**: Lithium carbonate prices will stabilize near current levels in a range between $20,000/mt and $25,000/mt from 2024 to 2027
- **Price Stability Importance**: Stability is key to attracting investment, with prices needing to be low enough for battery manufacturers to profit, but high enough to encourage suppliers toward continued expansion
- **Temporary Nature**: Some analysts suggest the oversupply may be temporary due to the rapid increase in demand expected in the coming years

Regional Dynamics

- **China's Dominance**: China has the highest production capacity for lithium-ion batteries, and its dominance is expected to continue despite trade barriers with the US and Europe
- **New Suppliers**: Lithium projects in Chile are progressing, and new suppliers like Brazil and Zimbabwe are emerging
- **US Push**: America's push for domestic production is ongoing but has not yet significantly altered the global market structure

출처: 1. https://www.bradley.com/insights/publications/2024/02/lithium-prices-in-free-fall-implications-for-clean-energy-transition-in-the-private-sector 2. https://www.sciencedirect.com/science/article/pii/S2949790625001004 3. https://www.fastmarkets.com/insights/facing-the-tightening-lithium-supply-challenge-in-2025/ 4. https://acuitytrading.com/r commentary/what-the-lithium-oversupply-means-for-the-global-economy 5. https://www.oxfordenergy.org/wpcms/wp-content/uploads/2024/02/Insight-145-Lithium-Price-Volatility.pdf

North American Lithium Projects Status (2023-2024)

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North American Lithium Projects Status (2023-2024)

United States

Albemarle - Silver Peak (Nevada)

- **Current Status**: Operational - the only small lithium mine operating in the US today
- **Expansion Plans**: Aims to increase lithium carbonate production from 5,000-10,000 tonnes per year by 2025
- **Strategic Shift**: Part of Albemarle's proactive measures to re-phase organic growth investments and optimize cost structure in response to changing market conditions
- **Capital Expenditure**: The company expects its 2024 capital expenditures to range from $1.6 billion to $1.8 billion, down from about $2.1 billion in 2023

Albemarle - Kings Mountain (North Carolina)

- **Current Status**: In development/permitting phase
- **Strategic Shift**: Albemarle deferred spending on its lithium conversion facility project in South Carolina and redirected efforts towards permitting activities for the Kings Mountain lithium-spodumene mine resource

- **Supply Chain Integration**: After lithium is extracted, it is sent to Albemarle's Kings Mountain processing facility in North Carolina, creating a domestic supply loop

Lithium Americas - Thacker Pass (Nevada)

- **Current Status**: Under construction
- **Production Targets**:
  - Expected to produce 40,000 metric tons of lithium carbonate per year once it starts up in 2027
  - Enough for lithium-ion batteries in about 800,000 electric cars
  - Aims to eventually increase production to 80,000 tonnes per year
- **Funding**:
  - Received a conditional commitment from the US Department of Energy for a $2.3 billion loan
  - Even with the DOE loan and full GM investment, estimated to need an additional $266 million to fully fund the approximately $3 billion project
- **Legal Challenges**:
  - Survived initial lawsuits brought forth by environmental groups and tribal communities in November 2023
  - Environmental groups have now turned their attention to changing The Mining Law of 1872
  - The Fish and Wildlife Service announced a year-long study in February 2024 into a local snail at Thacker Pass that could be deemed endangered

Controlled Thermal Resources - Salton Sea (California)

- **Current Status**: Construction began in January 2024
- **Project Scope**: Lithium mine and geothermal power plant
- **Production Targets**:
  - Initially produce 25,000 metric tons of lithium
  - Potential for up to 175,000 metric tons once completed
- **Legal Challenges**: In March 2024, sued by conservation and environmental justice groups for alleged violation of county and state laws based on a "fatally flawed" environmental review and lack of meaningful tribal consultation
- **Response**: Both CTR and Imperial County claim the process was properly conducted and that they comply with CEQA (California Environmental Quality Act)

Canada

Whabouchi Mine (Quebec)

- **Current Status**: In development
- **Location**: James Bay region, Quebec
- **Production Targets**: Expected to produce 235,000 metric tons of spodumene concentrate per year
- **Product**: The Whabouchi Mine will produce spodumene concentrate, which may be converted to lithium hydroxide
- **Ownership**: Part of Arcadium Lithium's portfolio

Overall Canadian Lithium Sector

- **Historical Production**: From 2014 to 2020, Canada's lithium production was limited
- **Current Resources**: Canada currently has an estimated 3.2 million tonnes of lithium oxide resources in hard rock deposits
- **Development Focus**: Several companies are advancing lithium projects across Canada, exploring both traditional hard rock mining and unconventional sources such as oilfield brines and industrial wastewaters
- **Government Support**: The Government of Canada has identified lithium as a critical mineral due to its importance in the renewable energy transition
- **Trade Status**: Canada is a net importer of lithium and lithium products
- **Investment**: Australia and South Korea have emerged as the largest investors in Quebec's critical minerals extraction and EV battery production

출처: 1. https://carboncredits.com/albemarle-shifts-focus-in-lithium-strategy-amid-market-softening/ 2. https://www.forbes.c could-be-the-year-for-american-lithium/ 3. https://cen.acs.org/energy/energy-storage-/Lithium-Americas-expects-23-billion/102/web/2024/03 4. https://natural-resources.canada.ca/minerals-mining/mining-data-statistics-analysis/minerals-metals-facts/lithium-facts 5. https://arcadiumlithium.com/wp-content/uploads/2024/09/2023-Arcadium-Lithium-Sustainability-Report.pdf

African Lithium Projects Status (2023-2024)

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African Lithium Projects Status (2023-2024)

Zimbabwe

Overview

- **Global Position**: Zimbabwe ranks sixth in lithium production globally (700 tonnes) and is poised to become a significant player
- **Extraction Method**: Lithium is extracted from petalite mineral found in pegmatites
- **Major Deposits**: Reserves concentrated at the Bikita, Zulu, and Kamativi mines
- **Challenges**: Economic extraction, marketing, environmental constraints in semi-arid areas (particularly water-related issues), and logistical difficulties due to being a landlocked nation

Bikita Minerals

- **Ownership**: Owned by Sinomine Resource Group (Chinese firm)
- **Recent Developments**: In 2024, reduced production and cut jobs
- **Cited Reasons**: Poor infrastructure, inconsistent policies on licensing, taxation and export regulations, as well as fluctuating global lithium prices

Policy Developments

- **Export Ban**: Zimbabwe has announced plans to ban the export of lithium concentrates from 2027 to boost local refining and increase revenues
- **Chinese Dominance**: Chinese firms have established a monopoly in Zimbabwe's lithium sector, often undermining local beneficiation initiatives
- **Future Vision**: Zimbabwe aims to focus on downstream industries such as lithium-ion battery manufacturing and electric vehicle production

Namibia

Karibib Project

- **Status**: Most advanced lithium project in the country, currently at the development stage
- **Production Timeline**: Expected to start in the second half of 2025
- **Scale**: Covers approximately 68 km^2

Uis Lithium Project

- **Scale**: Covers approximately 308 km^2
- **Challenges**: Minimal capacity for lithium mineral beneficiation, processing, and production, leading to export of mineral concentrate with value added elsewhere

Mali

Goulamina Project

- **Status**: Construction has started
- **Production Timeline**: Expected to come online in the first half of 2024
- **Significance**: Set to be West Africa's first spodumene producer

- **Scale**: Covers approximately 100 km^2

Ghana

Ewoyaa Project

- **Recent Developments**: Recently granted a 15-year mining permit by the Ghanaian Ministry of Lands and Natural Resources
- **Portfolio**: Part of the Cape Coast Lithium Portfolio, which includes the Ewoyaa Lithium Project and Egyasimanku Hill Lithium Project
- **Scale**: Covers approximately 509 km^2

Democratic Republic of Congo (DRC)

Manono Project

- **Scale**: Covers approximately 188 km^2
- **Regional Significance**: Part of the "African lithium triangle" along with Zimbabwe and Namibia

Regional Dynamics

- **African Lithium Triangle**: Zimbabwe, Namibia, and the DRC form the core of Africa's lithium resources
- **Economic Block**: These three nations already belong to the same economic block (SADCC), which makes collaboration more likely
- **Leading Nations**: Zimbabwe, Namibia, DRC, Mali, and Ghana form the African-leading lithium nations
- **Environmental Concerns**: Like the South American lithium triangle, the African lithium triangle is also threatened with environmental and ethical challenges
- **Water Limitations**: Extraction and processing of lithium in countries like Namibia and Zimbabwe is water-limited, similar to Chile

출처: 1. https://www.sciencedirect.com/science/article/pii/S0301420724004136 2. https://futures.issafrica.org/blog/2025/Zi lithium-beneficiation-policy-a-catalyst-for-Vision-2030 3. https://www.sciencedirect.com/science/article/pii/S0301420724001 4. https://www.spglobal.com/commodity-insights/en/news-research/latest-news/metals/061125-zimbabwe-to-ban-lithium-concentrate-exports-from-2027-to-boost-refining

Comprehensive Summary: Global Lithium Project Production Cuts, Closures, and Subsidies (2015-2024)

카테고리: summary

수집 시간: 2025-08-07 16:23:46

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Comprehensive Summary: Global Lithium Project Production Cuts, Closures, and Subsidies (2015-2024)

Market Context and Price Dynamics

The global lithium market experienced extreme volatility between 2015 and 2024, with prices reaching unprecedented highs before collapsing dramatically. After spot prices peaked at over $80,000 per ton in December 2022, they plummeted to just over $13,000 per ton by January 2024—a decline of more than 80%. This price crash was driven by significant oversupply in the market, coupled with softer-than-expected demand, particularly from the electric vehicle (EV) sector.

Global lithium production expanded rapidly, jumping from approximately 737,000 tonnes in 2022 to almost 1.2 million tonnes in 2024 on a lithium carbonate equivalent (LCE) basis. This led to substantial market surpluses—estimated at

around 175,000 tonnes in 2023 and 154,000 tonnes in 2024. The oversupply situation was exacerbated by slower-than-anticipated EV adoption rates, with the EV penetration rate in the US actually declining from 7.5% in 2023 to 6.8% in the first half of 2024.

Production Cuts and Closures by Region

Australia

Australia, as the world's largest lithium producer, experienced the most significant production cuts in response to falling prices:

1. **Core Lithium - Finniss Project**: Operations were suspended in January 2024 due to sustained lithium price declines. The project, which was designed for 197,000 tonnes of annual spodumene production with an eight-year mine life, is now only processing stockpiles with final sales expected in Q3 2024.
2. **Pilbara Minerals - Ngungaju Plant**: Placed into care and maintenance in 2024, with approximately 50 workers shed as part of the decision to mothball the plant. The Ngungaju facility had a capacity of 200,000 tonnes.
3. **Talison Lithium - Greenbushes Mine**: Implemented a production cut of approximately 100,000 tonnes. Before the cuts, the mine had a capacity of 1.5 million tonnes per annum and was undergoing expansion to 2 million tonnes.
4. **Liontown Resources - Kathleen Valley**: Deferred expansion and ramp-up of its project as part of a cost-cutting review. The project began production in 2024 with a planned capacity of 500,000 tonnes per annum.

Overall, Australian production in Q1 2024 declined by about 12.3% compared to the previous quarter, with production cuts starting in January 2024 but building momentum throughout the year.

China

China's response to the price crash differed significantly from Western producers:

1. **Strategic Overproduction**: While non-Chinese companies pulled back production volume significantly, Chinese-owned companies pulled back much less and in some cases continued to expand production, generating an oversupply of lithium and other battery materials.
2. **Government Support**: Chinese government policies enabled companies to sell materials at low prices that would otherwise be unsustainable, maintaining market share rather than balancing supply with demand.
3. **Lepidolite Mine Closure**: A major Chinese producer finally halted output at one of the biggest lepidolite mines in September 2024, causing three associated lepidolite processing plants to shut down. This was one of the first significant production cuts in China, occurring later than cuts in Western countries.

Despite some cuts, China's lithium supply grew by nearly 15% year-on-year, from 35,700 metric tons in 2023 to 41,000 metric tons in 2024, maintaining its position as the third-largest lithium producer globally.

South America

South American lithium producers, particularly in Chile and Argentina, focused more on expansion than cuts during this period:

1. **SQM (Chile)**: Rather than cutting production, SQM announced plans to expand production of lithium hydroxide from 35,000 metric tons to 100,000 metric tons by the end of 2024, and to increase lithium carbonate production from 185,000 metric tons to 240,000 metric tons by 2026.
2. **Albemarle (Chile)**: Continued operations at its Atacama and La Negra facilities without significant cuts.
3. **Argentina**: Achieved an 87.5% increase in annual production in 2024, producing 33,000 tonnes of LCE in 2022, making it the second-largest producer in the region and fourth globally.
4. **Chilean Government Policy**: In April 2023, Chile announced plans for nationalizing its lithium industry, primarily affecting SQM and Albemarle. This led to new partnerships, such as the preliminary public-private deal between SQM and state-owned Codelco signed in May 2024.

North America

North American lithium projects were largely in development rather than production during this period, so they experienced strategic shifts rather than outright cuts:

1. **Albemarle - Silver Peak (Nevada)**: The only operational lithium mine in the US maintained production but adjusted its expansion plans, aiming to increase lithium carbonate production from 5,000-10,000 tonnes per year by 2025.
2. **Albemarle - Strategic Shift**: The company deferred spending on its lithium conversion facility project in South Carolina and redirected efforts towards permitting activities for the Kings Mountain lithium-spodumene mine resource in North Carolina. It reduced 2024 capital expenditures to $1.6-1.8 billion, down from about $2.1 billion in 2023.
3. **New Projects Under Development**: Several major projects began construction, including Lithium Americas' Thacker Pass (Nevada) and Controlled Thermal Resources' Salton Sea (California) project, though both faced legal challenges.

Africa

African lithium projects were mostly in development stages during this period, with some production adjustments:

1. **Bikita Minerals (Zimbabwe)**: Owned by Sinomine Resource Group, reduced production and cut jobs in 2024, citing poor infrastructure, inconsistent policies, and fluctuating global lithium prices.
2. **Projects in Development**: Several major projects were in development, including the Karibib Project in Namibia (expected to start production in the second half of 2025), the Goulamina Project in Mali (expected to come online in the first half of 2024), and the Ewoyaa Project in Ghana (recently granted a 15-year mining permit).

Government Subsidies and Support

United States

1. **Inflation Reduction Act (IRA)**: Established critical mineral requirements for EV tax credits, with increasing percentages of battery minerals (including lithium) required to be sourced from the US or free trade agreement countries—50% in 2023, rising to 100% by 2029.
2. **Department of Energy Loans**: Provided significant financial support for lithium projects, including a conditional $2.3 billion loan to Lithium Americas for its Thacker Pass project.
3. **Processing Capacity Focus**: The US government prioritized expansion of processing capacity for lithium and other battery materials, with tariffs and tax incentives resulting in favorable pricing for lithium project developers.

China

1. **Strategic Overproduction Support**: Government policies enabled Chinese companies to maintain production despite low prices, supporting a strategy of market dominance.
2. **EV Subsidy Expiration**: The Chinese Government's decade-long program of subsidies for electric vehicle purchases expired in 2023, contributing to price volatility.
3. **Global Acquisition Strategy**: Chinese companies bought half of all lithium mines offered for sale since 2018, in addition to domestically mining 18% of global lithium, with government backing for these acquisitions.

Australia

1. **Critical Minerals Strategy**: The Australian Government's Critical Minerals Strategy (2023) emphasized developing domestic resources to support both national interests and global clean energy transitions.
2. **Environmental Standards**: The strategy noted that "Critical minerals projects that demonstrate superior environmental and social practices are increasingly favored by both regulators and investors."

Other Regions

1. **Chile's Nationalization**: Chile's move toward nationalizing its lithium industry represented a shift from subsidizing private companies to creating state-controlled enterprises to capture more value from lithium resources.

2. **Zimbabwe's Beneficiation Policy**: Zimbabwe announced plans to ban the export of lithium concentrates from 2027 to boost local refining and increase revenues, aiming to develop downstream industries.

Conclusion

The period from 2015 to 2024 saw the global lithium industry transition from boom to bust and the beginnings of a potential rebalancing. Production cuts were most pronounced in high-cost operations, particularly in Australia, while Chinese producers maintained output despite falling prices. Government policies played a crucial role in shaping the industry, with the US and other Western nations focusing on securing supply chains through subsidies and domestic production incentives, while China pursued a strategy of market dominance through overproduction and strategic acquisitions.

The production cuts and project delays announced in 2023-2024, combined with continued growth in demand for lithium batteries, are expected to eventually rebalance the market. Analysts project that lithium carbonate prices will stabilize in a range between $20,000-$25,000 per metric ton from 2024 to 2027, providing a foundation for more sustainable industry growth.

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Research Summary: Lithium Project Production Cuts, Closures, and Subsidies (2015-2024)

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Research Summary: Lithium Project Production Cuts, Closures, and Subsidies (2015-2024)

☒ 최신 정보: 2024년 리튬 시장 및 프로젝트 현황

This comprehensive research examines lithium project production cuts, closures, and government subsidies across global markets from 2015 to 2024, with particular focus on the dramatic market shifts of 2023-2024.

Key Findings

Market Context

- **Price Collapse**: Lithium prices fell by over 80% from peak levels of $80,000/ton in December 2022 to approximately $13,000/ton by January 2024
- **Supply Surplus**: Global market recorded surpluses of approximately 175,000 tonnes in 2023 and 154,000 tonnes in 2024
- **Production Growth**: Global lithium production expanded from 737,000 tonnes in 2022 to nearly 1.2 million tonnes in 2024
- **Demand Shortfall**: Slower-than-expected EV adoption contributed to oversupply, with US EV penetration declining from 7.5% in 2023 to 6.8% in early 2024

Major Production Cuts and Closures

Australia (Most Significant Cuts)

1. **Core Lithium - Finniss Project**

    - Operations suspended January 2024
    - Designed capacity: 197,000 tonnes annual spodumene production
    - Current status: Processing stockpiles only, final sales expected Q3 2024
2. **Pilbara Minerals - Ngungaju Plant**
    - Placed into care and maintenance in 2024
    - Capacity: 200,000 tonnes
    - Workforce impact: 50 workers shed, 50 redeployed
3. **Talison Lithium - Greenbushes Mine**
    - Production cut of approximately 100,000 tonnes
    - Previous capacity: 1.5 million tonnes per annum
    - Ownership: Albemarle
4. **Liontown Resources - Kathleen Valley**
    - Expansion and ramp-up deferred
    - Planned capacity: 500,000 tonnes per annum
    - Began production in 2024 before cuts announced

China (Strategic Overproduction)

1. **Lepidolite Mine Closure**
    - Major producer halted output September 2024
    - Impact: Three associated processing plants shut down
    - First significant Chinese production cut, much later than Western cuts
2. **Overall Strategy**
    - Chinese companies maintained or expanded production despite price crash
    - Government policies enabled selling at otherwise unsustainable prices
    - Lithium supply grew 15% year-on-year (35,700 to 41,000 metric tons) despite market conditions

South America (Focus on Expansion)

1. **SQM (Chile)**
    - Expansion plans: Increase lithium hydroxide from 35,000 to 100,000 metric tons by end of 2024
    - Further plans: Increase lithium carbonate from 185,000 to 240,000 metric tons by 2026
    - Government relations: New partnership with state-owned Codelco signed May 2024
2. **Argentina**
    - Production growth: 87.5% increase in annual production in 2024
    - 2022 production: 33,000 tonnes of LCE

North America (Strategic Shifts)

1. **Albemarle - Silver Peak (Nevada)**
    - Status: Only operational US lithium mine
    - Adjusted expansion: Aims to increase production from 5,000 to 10,000 tonnes/year by 2025
    - Capital expenditure: Reduced 2024 capex to $1.6-1.8 billion from $2.1 billion in 2023
2. **New Projects**
    - Lithium Americas (Thacker Pass): Construction underway, targeting 40,000 tonnes/year by 2027
    - Controlled Thermal Resources (Salton Sea): Construction began January 2024, targeting 25,000 tonnes initially

Africa (Emerging Production)

1. **Bikita Minerals (Zimbabwe)**
    - Reduced production and cut jobs in 2024
    - Cited reasons: Poor infrastructure, inconsistent policies, fluctuating prices
    - Ownership: Sinomine Resource Group (Chinese)

Government Subsidies and Support

United States

1. **Inflation Reduction Act**
    - Critical mineral requirements for EV tax credits

    - Percentage requirements: 50% in 2023, increasing to 100% by 2029
    - Investment impact: Generated over $224 billion in clean energy investments
2. **Direct Project Support**
    - $2.3 billion conditional loan to Lithium Americas for Thacker Pass
    - Processing capacity prioritization for lithium, graphite, cobalt, and nickel

China

1. **Strategic Support**
    - Enabled overproduction despite low prices
    - Facilitated acquisition of global lithium assets (50% of mines offered for sale since 2018)
    - EV subsidy program expired in 2023 after decade-long implementation

Other Regions

1. **Chile**: Nationalization plans announced April 2023, shifting from private to state-controlled production
2. **Zimbabwe**: Plans to ban lithium concentrate exports from 2027 to boost domestic refining
3. **Australia**: Critical Minerals Strategy (2023) emphasizing environmental and social practices

Regional Production Impact Analysis

Australia

- Q1 2024 production declined 12.3% compared to previous quarter
- Production cuts started January 2024 but accelerated throughout the year
- High-cost producers most affected by price decline

China

- Maintained dominant market position despite global downturn
- End of 2024: Lithium salt inventories at 220,000 tonnes (20% of annual demand)
- Post-holiday restocking failed due to slow downstream buying activity

South America

- Chile expanded output 80% since 2016 to about 140,000 tons annually (as of 2022)
- All Chilean lithium extracted from Salar de Atacama as of 2023
- Bolivia's production limited to 600 tonnes of LCE in 2023

Future Outlook

- Analysts project lithium carbonate prices to stabilize between $20,000-$25,000/mt from 2024 to 2027
- Production cuts and project delays expected to eventually rebalance the market
- Some analysts suggest oversupply may be temporary due to projected demand growth
- Global lithium-ion battery market expected to remain oversupplied until 2028

This research provides a comprehensive overview of how the lithium market evolved from 2015-2024, with particular emphasis on the production cuts, closures, and government interventions that shaped the industry during this volatile period.

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Salar de Atacama (SQM and Albemarle) Lithium Operations 2015-2024

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Salar de Atacama Lithium Operations (Chile)

Basic Information

- **Region**: South America
- **Country**: Chile
- **Operators**: SQM and Albemarle
- **Status**: Operational
- **Start Date**: Pre-2015
- **Significance**: Salar de Atacama is the world's largest and richest lithium resource, containing almost half the world's known reserves according to USGS.

Production Information

- SQM's Salar de Atacama was the top-producing lithium brine operation globally in 2023, with production of 166,000 metric tons of lithium carbonate.
- Albemarle produces lithium carbonate at its La Negra lithium conversion plants, which process brine from the Salar de Atacama.

Production Cuts (2015-2024)

- No specific production cuts were identified for the Salar de Atacama operations between 2015-2024. Instead, both SQM and Albemarle have been expanding production during this period.
- In January 2024, Albemarle announced plans to cut staff and pause expansions amid falling lithium prices, but this did not specifically target the Salar de Atacama operations.

Expansion Plans (2023-2024)

- SQM has been expanding its lithium carbonate capacity in Chile. It expects to produce 210,000 tons in 2024 and 240,000 tons by 2025.
- SQM plans to expand production of lithium hydroxide from 35,000 metric tons to 100,000 metric tons by the end of 2024.
- Albemarle aims to implement direct lithium extraction technology at the salt flat to reduce water usage.

Financial Performance

- In Q1 2024, SQM posted a loss of $870 million, compared to a profit of $138 million in Q1 2025.
- While SQM's quarterly lithium revenues fell 67% from a year earlier, its lithium sales volumes grew 34%.

- SQM predicted volumes could climb some 18% to reach 200,000 metric tons in 2024, compared to 170,000 tons in 2023.

Government Policies and Agreements

- In April 2023, Chilean President Gabriel Boric announced that private companies will need to partner with the state to extract lithium.
- In December 2023, SQM reached a framework agreement to hand over a majority stake in its Chilean brine assets to state-owned Codelco in exchange for extending mining rights.
- In May 2024, SQM signed a preliminary public-private deal with Chile's state-owned copper mining company Codelco.

Environmental Concerns

- A recent study showed that lithium mining is slowly sinking Chile's Atacama salt flat.
- In 2016, Chile's SMA environmental regulator charged SQM with overdrawing lithium-rich brine from the Salar de Atacama salt flat, prompting the company to develop a $25 million plan to address the issue.

Market Outlook

- SQM believes that strong demand growth in the lithium market seen since the beginning of 2024 could continue for the remainder of the year, with total lithium demand surpassing 1.1 million metric tons during 2024.

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Australian Lithium Projects (Pilgangoora, Mt Marion, Kathleen Valley) 2015-2024

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Australian Lithium Projects (2015-2024)

1. Pilgangoora Project (Pilbara Minerals)

Basic Information

- **Region**: Oceania
- **Country**: Australia
- **Operator**: Pilbara Minerals
- **Status**: Operating with partial closure
- **Start Date**: 2018
- **Production Capacity**: 1,000,000 tonnes per annum

Production Cuts and Closures

- **October 2024**: Pilbara Minerals announced the suspension of its Ngungaju processing plant and cut lithium output.
- **2024**: The company shed about 50 workers as part of its decision to mothball the plant, and redeployed the remaining 50 employees from the facility across its operations.
- **Reason for Cuts**: Market oversupply and falling lithium prices.

- **Quote**: "With the market oversupplied production cuts are needed, so Pilbara's decision to idle the Ngungaju plant is an important development," according to William Adams, Fastmarkets head of base metal and battery raw materials research.

Production History

- **FY2024 production**: 725,329 tonnes of spodumene concentrate
- **Ngungaju Plant**: Began production in 2018, was purchased from bankruptcy and restarted in 2021, then placed into care and maintenance in 2024 due to low prices. Capacity: 200,000 tonnes.

2. Mt Marion Lithium Project

Basic Information

- **Region**: Oceania
- **Country**: Australia
- **Operators**: Mineral Resources and Ganfeng Lithium Group
- **Status**: Operating
- **Start Date**: 2017
- **Production Capacity**: 900,000 tonnes per annum (after 2023 upgrades)

Production Information

- **2023 Production**: Estimated 46.6 thousand tonnes of lithium
- **Q1 2025**: Mt Marion produced 140,000 tonnes of lithium concentrate, up 21% quarter-on-quarter but down 23% year-on-year, with an average product grade of 4.3%.
- **FY2025**: Mt Marion raised its production target by 70,000 tonnes
- **Future Plans**: As of June 2024, underground development is underway at Mount Marion with first production from this new phase in the mine's operations beginning by the end of 2025.

Business Arrangements

- Mineral Resources secured its first offtake partner in 2015 when Ganfeng Lithium invested AUD$135 million and committed to purchase 100% of the project's production.
- In 2022, Mineral Resources exercised its option to take back 51% of the offtake.
- In February 2023, Mineral Resources signed an agreement to increase its ownership from 40% to 50%, with Ganfeng owning the remaining interest.

Production Cuts

- No specific production cuts were identified for Mt Marion between 2015-2024. Instead, production has been increasing.

3. Kathleen Valley Lithium Project

Basic Information

- **Region**: Oceania
- **Country**: Australia
- **Operator**: Liontown Resources
- **Status**: Operating (newly commissioned)
- **Start Date**: July 2024
- **Production Capacity**: 500,000 tonnes per annum (planned to increase to 700,000 tonnes in year six)

Production Information

- **Q1 FY2025**: Kathleen Valley produced 95,709 tonnes of spodumene concentrate, up 12% quarter-over-quarter.
- **Expected Production**: Approximately 500,000 MT per year of spodumene concentrate by the end of Q1 2025.
- **Future Plans**: An expansion planned in year six to bring its production to 700,000 MT per year.

Development History

- In January 2023, plant design optimization increased processing capacity by 20%, enabling a nameplate throughput of 3 million tonnes per year at start-up (Q2 2024).
- The project plans to ramp up further by Year 6 to 4 million tonnes per year of ore throughput.

Production Cuts

- No production cuts have been reported for Kathleen Valley as it only began operations in July 2024.

Market Context for Australian Lithium Production

- **2019**: Wodgina mine (operated by Mineral Resources) closed in 2019 due to low prices, but restarted production in 2022.
- **2024**: Bald Hill mine (operated by Mineral Resources) was placed into care and maintenance in 2024 due to low prices, with the loss of up to 300 jobs.
- **November 2024**: Albemarle chief executive Kent Masters said when announcing a $1 billion third quarter loss that he expected lithium to stay in a range of $12 to $15/kg for the foreseeable future.
- **2024**: Another Mineral Resources lithium mine run in joint venture with Albemarle Corporation was on notice that it might also close or be severely scaled back.

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Chinese Lithium Projects (Chaerhan Lake, Yiliping, East Taijnar) 2015-2024

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Chinese Lithium Projects (2015-2024)

1. Chaerhan Lake Mine

Basic Information

- **Region**: Asia
- **Country**: China
- **Location**: Qinghai Province
- **Operator**: Qinghai Salt Lake Industry
- **Status**: Operating
- **Type**: In-situ leaching mine

Production Information

- **2023 Production**: Estimated 27.5 thousand tonnes of lithium
- **2024 Projected Production**: 52,000 tons of LCE
- **2025 Projected Production**: 82,000 tons of LCE

Production Trends

- The total production from Chaerhan Salt Lake is on an upward trajectory, showing robust growth.
- No specific production cuts were identified for this project between 2015-2024.

2. East Taijinar Lake

Basic Information

- **Region**: Asia
- **Country**: China
- **Operator**: Western Mining Co., Ltd.
- **Status**: Operating
- **Production Capacity**: 20,000 tons of LCE per year

Production Plans

- The company has planned expansion to add an additional 10,000 tons to the annual production capacity.
- No specific production cuts were identified for this project between 2015-2024.

3. Liping Salt Lake (Yiliping)

Basic Information

- **Region**: Asia
- **Country**: China
- **Operator**: China Minmetals Corporation
- **Status**: Operating
- **Production Capacity**: 10,000 tons of lithium carbonate equivalent (LCE) per year

Production Information

- **2023 Production**: 14,400 tons LCE, exceeding the stated capacity by a substantial margin
- The overproduction suggests efficient optimization of resources and possible expansion in operations.
- No specific production cuts were identified for this project between 2015-2024.

Chinese Lithium Industry Production Cuts (2022-2024)

General Industry Trends

- **September 2023**: A major Chinese producer halted output at one of the biggest lepidolite mines in the country, causing three associated lepidolite processing plants to shut down.
- **Q4 2024**: Further production cut announcements emerged, along with more project start-up delays and curtailments.
- **2024**: Chinese producers were not immune to weakening prices in 2024, with one of China's largest lepidolite mines halting production due to its higher production costs.
- **September 2024**: A Chinese lithium company issued a notice that its holding subsidiary company would carry out a 15-day production suspension and maintenance work, while its wholly-owned subsidiary would carry out a 10-day production suspension and maintenance work.

Market Context

- **2023-2024**: The lithium market recorded a surplus of around 175,000 tonnes in 2023, and almost 154,000 tonnes in 2024 based on available data.
- **2024**: Prices of lithium carbonate minimum 99.5% CIF North Asia were seen at $12,627/t over 2024, down 82% from 2022 levels.
- **2024**: Global lithium chemical supply was expected to remain around 9.2% higher than consumption in 2024.
- **2024**: Buying interest for lithium in China remained thin, with most market participants awaiting cues on output and prices.

Chinese Government Subsidies and Support (2015-2024)

Direct Subsidies to Battery Manufacturers

- **2022**: The Chinese government provided $401 million in subsidies to CATL (China's largest EV battery maker).
- **2023**: Subsidies to CATL increased to $809 million (more than double the 2022 amount).
- **2023**: The government provided $208.9 million to EVE Energy (China's fourth-largest EV battery producer).
- **2018-2023**: The Chinese government extended a total of $1.8 billion in subsidies to CATL alone.

Policy Support

- **2009-2017**: Buyer's rebates accounted for 62% of the Chinese government's support for the EV sector.
- **2022**: The Chinese government reduced the buyer's rebate.
- **2015-2024**: The Chinese government implemented a subsidy-centric policy that introduced fresh funds for mining, resources exploration, and processing machinery.
- **2015-2024**: Subsidies were reserved for domestic battery producers, which marked a turning point for the battery industry in China.
- **2024-2027**: China extended its tax exemption policy for New Energy Vehicles (NEVs) until 2027 to promote the electric vehicle industry.

Strategic Investments

- The Chinese government aggressively pursued the acquisition of materials crucial for the global green energy transition.
- Chinese firms have made significant investments in lithium-rich countries like the "Lithium Triangle" (Argentina, Chile, and Bolivia) to further vertically integrate the supply chain for lithium-ion batteries.

Export Performance

- **2024**: China's lithium-powered battery exports surpassed $60 billion.
- **December 2024**: Exports to the United States hit an all-time high of nearly $2 billion in monthly sales.
- **2023**: Europe became China's largest market for EVs, with almost 40% of China's 2023 EV exports going to Europe.

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Argentinian Lithium Projects (Olaroz, Salar del Hombre Muerto) 2015-2024

카테고리: market_analysis

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Argentinian Lithium Projects (2015-2024)

1. Olaroz Project

Basic Information

- **Region**: South America
- **Country**: Argentina
- **Province**: Jujuy
- **Location**: Salar de Olaroz
- **Operator**: Sales de Jujuy (Orocobre 67.5%, Toyota Tsusho 25%, JEMSE 8.5%)
- **Status**: Production
- **Start Date**: Pre-2015

Ownership Structure

- Public-Private Partnership:
  - Orocobre (Australia): 66.5%

- Toyota Tsusho Corporation (Japan): 25%
- JEMSE (Argentina): 8.5% ownership

Production Information

- The project has been in production since before 2015.
- In August 2021, Orocobre Ltd merged with Galaxy Resources Ltd to create Allkem.

Production Cuts and Disruptions

- **2020**: The COVID-19 pandemic interrupted lithium operations in Argentina, including at Olaroz.
- According to Reuters (April 8, 2020), some lithium projects in Argentina were halted due to the coronavirus crisis.
- No specific permanent closures were identified for this project between 2015-2024.

2. Salar del Hombre Muerto (Mina Fénix/Phoenix Project)

Basic Information

- **Region**: South America
- **Country**: Argentina
- **Province**: Catamarca
- **Operator**: Minera del Altiplano (Livent)
- **Status**: Production
- **Start Date**: Pre-2015

Ownership Structure

- 100% private (United States) - FMC Lithium Affiliate (now Livent)

Production Information

- The project has been in production since before 2015.
- Located in the Salar del Hombre Muerto, a high-altitude wetland more than 4,000 meters above sea level.

Legal and Environmental Issues

- **March 14, 2024**: An Argentine court suspended the issuance of new lithium mining permits in the area of the Salar del Hombre Muerto, recognizing the catastrophic impacts on local water sources, the environment, and the livelihoods of local communities.
- The ruling did not affect existing operations but prevented new permits.
- In the Salar del Hombre Muerto alone, there are eight lithium mining projects.

3. Sal de Vida Project

Basic Information

- **Region**: South America
- **Country**: Argentina
- **Province**: Catamarca and Salta
- **Location**: Salar del Hombre Muerto
- **Operator**: Originally Galaxy Resources Ltd, now Allkem (after merger with Orocobre Ltd in August 2021)
- **Status**: Development
- **Planned Production**: 600,000 tons of lithium carbonate over 40 years

Project Details

- The project plans to extract brine from the Salar through eight pumping wells.
- Once fully operational, it will produce up to 600,000 tons of lithium carbonate for export to Europe and the United States.
- The project includes an aqueduct to extract water from the river Los Patos for processing.
- Estimated average annual production: Stage 1: 40,000 tons/year LCE, Stage 2: expansion of at least 20,000 tons/year LCE.

- Estimated mine life: 40 years.
- Mining Method: Pumping - Chemical adsorption (Direct Lithium Extraction - DLE).

4. Hombre Muerto West Project

Basic Information

- **Region**: South America
- **Country**: Argentina
- **Location**: Salar del Hombre Muerto
- **Status**: Development
- **Reserves (2024)**: 806.4 kt LCE with average grade of 864.2 mg/l

Project Details

- The project is in the field west and Santa Bárbara areas.
- A Preliminary Economic Assessment was completed in March 2024.

Market Context for Argentinian Lithium Production

Government Policies

- The Mining Code, last modified in 1997 during President Menem's neoliberal era (1989-1999), explicitly established the concessionable nature of lithium, whose mines can only be exploited after concessions granted by the provincial governments.
- This context has proven adverse in disputed border territories, as between Salta and Catamarca, both of which have historical claims over lands in the Salar de Hombre Muerto.

Environmental and Social Concerns

- Lithium mining in Argentina's salt flats has raised significant environmental concerns, particularly regarding water usage in arid regions.
- Indigenous communities and local settlements have expressed opposition to some projects due to concerns about water resources and environmental impacts.
- The March 2024 court ruling to suspend new permits in the Salar del Hombre Muerto area highlights the growing legal challenges to lithium mining expansion in Argentina.

Production Outlook

- Despite the pandemic-related interruptions in 2020, Argentina's lithium industry was expected to recover relatively fast, encouraged by growing global commitment to electric vehicles.
- As of 2024, Argentina continues to develop its lithium sector with multiple projects in various stages of development.
- No major production cuts or permanent closures were identified for the main Argentinian lithium projects between 2015-2024, though temporary disruptions occurred during the COVID-19 pandemic.

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Lithium Mining Subsidies and Government Support (2015-2024)

카테고리: policy_research

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Lithium Mining Subsidies and Government Support (2015-2024)

Australia

Western Australia Lithium Industry Support Program (2024)

- **Program Value**: Total value of AU$99.37 million
- **Announced**: November 2024
- **Key Components**:
  - Government Trading Enterprises (GTEs) including DevelopmentWA, Synergy, and Water Corporation will temporarily waive government fees to support the continuation of downstream processing of lithium for up to 2 years, to a total value of AU$90 million.
  - For lithium miners in the ramp-up phase of their project, port charges and mining tenement fees will be waived for up to 24 months, to the value of AU$9.37 million.
- **Rationale**: To protect critical Western Australian jobs and support the state's push to become a global battery minerals processing hub.
- **Industry Significance**: In the 2023-24 financial year, lithium miners in Western Australia generated AU$8.4 billion in sales and delivered AU$710 million in State royalties. During the same period, the industry supported more than 11,000 Western Australian jobs.

Australian Critical Minerals Strategies (2019-2024)

- **Total Investment**: Through 17 critical minerals strategies since 2019, Australian governments have dedicated AU$6.6 billion for critical mineral developments.
- **Global Position**: Australia led global lithium production in 2023, with five of its mines contributing more than half of the world's total output.
- **Geopolitical Context**: Australia has emerged as a critical country of interest for the United States due to its abundance of critical minerals, particularly lithium. The Biden administration has promised to extend the application of its subsidies to 'friendly' and 'like-minded' countries, including Australia.

United States

Inflation Reduction Act (IRA) Provisions for Critical Minerals

- **Implementation**: The US has extended the application of its subsidies to 'friendly' and 'like-minded' countries, with Australia emerging as a critical country of interest due to its abundance of critical minerals.
- **Geopolitical Strategy**: The US is targeting China's involvement in Australian critical minerals as part of its strategy to de-risk its supply chains from China's dominance.
- **Impact on Investment**: Some analysts have argued that while the IRA is strict towards many Australia-based companies with significant Chinese ownership, investors, or processing arrangements, tapping into the subsidies and grants offered by Washington will most likely involve setting up shop in the US.

Canada

North American Lithium Mine Case Study

- The North American Lithium mine near Val-d'Or, Quebec (formerly Quebec Lithium) commenced commercial production in early 2018, shipping spodumene concentrate to China for processing into lithium carbonate.
- Due to a sharp decline in lithium and spodumene concentrate prices later that year, the mine suspended operations in 2019 and sought bankruptcy protection.
- It was acquired by Sayona Quebec in 2021 in a joint venture between Sayona Mining (Australia) and Piedmont Lithium (United States).
- This case illustrates the volatility of the lithium market and the role of international investment in reviving suspended operations.

Critical Minerals Production

- Canada, along with Australia and the US, ranked as major critical mineral producers among the Five Eyes countries.
- From 2023 to 2030, Canada and Australia together are expected to represent 57% of planned refining projects for cobalt.

China

Direct Subsidies to Battery Manufacturers

- **2022**: The Chinese government provided $401 million in subsidies to CATL (China's largest EV battery maker).
- **2023**: Subsidies to CATL increased to $809 million (more than double the 2022 amount).
- **2023**: The government provided $208.9 million to EVE Energy (China's fourth-largest EV battery producer).
- **2018-2023**: The Chinese government extended a total of $1.8 billion in subsidies to CATL alone.

Policy Support for Electric Vehicles and Batteries

- **2009-2017**: Buyer's rebates accounted for 62% of the Chinese government's support for the EV sector.
- **2022**: The Chinese government reduced the buyer's rebate.
- **2015-2024**: The Chinese government implemented a subsidy-centric policy that introduced fresh funds for mining, resources exploration, and processing machinery.
- **2015-2024**: Subsidies were reserved for domestic battery producers, which marked a turning point for the battery industry in China.
- **2024-2027**: China extended its tax exemption policy for New Energy Vehicles (NEVs) until 2027 to promote the electric vehicle industry.

Global Context and Market Dynamics

Market Oversupply and Price Impacts

- The oversupply in the lithium market was primarily fueled by the development of spodumene mines in Australia—two in 2017 and five more in 2018.
- By the end of 2019, lithium prices had fallen to US$7,300 per tonne, and by the close of 2020, they had dropped below US$5,000 per tonne.
- This price decline led to production cuts and mine closures, particularly in Australia, as documented in the case of the North American Lithium mine in Canada.

Geopolitical Dimensions

- The increasingly binary geopolitical reality - China versus the US - is pushing Australia, which is rich in many critical minerals, particularly lithium, to the frontline of great power rivalry.
- This is forcing countries like Australia to apply a zero-sum geopolitical logic to their role in global supply chains, which often trumps economic rationality and green transition goals.
- Despite Chinese investment in Australia declining since 2013 and in Canada since 2019, these two countries remain among the top five destinations for Chinese State-Owned Enterprise acquisitions in the mining sectors from 2013 to 2023.

Processing Dominance

- China dominates lithium processing, with most of the processed products being used in battery manufacturing.
- This dominance in processing has led to strategic concerns in Western countries, resulting in increased government support for domestic processing capabilities.

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Summary of Lithium Projects Production Cuts, Closures, and Subsidies (2015-2024)

카테고리: summary

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Summary of Lithium Projects Production Cuts, Closures, and Subsidies (2015-2024)

Production Cuts and Closures

1. Chile (Salar de Atacama)

- **No Major Production Cuts**: Despite global lithium price fluctuations, the major operations at Salar de Atacama (SQM and Albemarle) did not implement significant production cuts between 2015-2024. Instead, both companies have been expanding production.
- **Expansion Plans**: SQM expects to produce 210,000 tons of lithium carbonate in 2024 and 240,000 tons by 2025. SQM also plans to expand production of lithium hydroxide from 35,000 metric tons to 100,000 metric tons by the end of 2024.
- **Financial Challenges**: In Q1 2024, SQM posted a loss of $870 million, compared to a profit of $138 million in Q1 2025, reflecting the impact of falling lithium prices.
- **Government Intervention**: In April 2023, Chilean President Gabriel Boric announced that private companies will need to partner with the state to extract lithium, leading to SQM reaching an agreement with state-owned Codelco in December 2023.

2. Australia

- **Pilgangoora Project (Pilbara Minerals)**:
  - **October 2024**: Announced the suspension of its Ngungaju processing plant and cut lithium output.
  - **Job Losses**: Shed about 50 workers as part of its decision to mothball the plant.
  - **Reason**: Market oversupply and falling lithium prices.
- **Wodgina Mine (Mineral Resources)**:
  - **2019**: Closed due to low lithium prices.
  - **2022**: Restarted production as prices recovered.
- **Bald Hill Mine (Mineral Resources)**:
  - **2024**: Placed into care and maintenance due to low prices, with the loss of up to 300 jobs.
- **Other Australian Operations**:
  - **2024**: Another Mineral Resources lithium mine run in joint venture with Albemarle Corporation was on notice that it might also close or be severely scaled back.
  - **Mt Marion and Kathleen Valley**: No production cuts were identified; instead, these operations have been increasing production.

3. China

- **September 2023**: A major Chinese producer halted output at one of the biggest lepidolite mines in the country, causing three associated lepidolite processing plants to shut down.
- **Q4 2024**: Further production cut announcements emerged, along with more project start-up delays and curtailments.
- **September 2024**: A Chinese lithium company issued a notice that its holding subsidiary company would carry out a 15-day production suspension and maintenance work.
- **Major Salt Lake Operations**: No specific production cuts were identified for Chaerhan Lake, East Taijinar Lake, or Liping Salt Lake (Yiliping) between 2015-2024. Instead, these operations have been expanding production.

4. Argentina

- **2020**: The COVID-19 pandemic interrupted lithium operations in Argentina, including at Olaroz and other projects.
- **No Permanent Closures**: No major permanent closures were identified for the main Argentinian lithium projects (Olaroz and Salar del Hombre Muerto) between 2015-2024.
- **Legal Challenges**: In March 2024, an Argentine court suspended the issuance of new lithium mining permits in the area of the Salar del Hombre Muerto due to environmental concerns, though this did not affect existing operations.

Government Subsidies and Support

1. Australia

- **Western Australia Lithium Industry Support Program (2024)**:
  - **Total Value**: AU$99.37 million
  - **Key Components**: Waiving of government fees for downstream processing (AU$90 million) and waiving of port charges and mining tenement fees for miners in the ramp-up phase (AU$9.37 million).
  - **Duration**: Up to 2 years
- **Critical Minerals Strategies (2019-2024)**:
  - **Total Investment**: AU$6.6 billion dedicated for critical mineral developments through 17 strategies since 2019.

2. China

- **Direct Subsidies to Battery Manufacturers**:
  - **2022**: $401 million to CATL
  - **2023**: $809 million to CATL and $208.9 million to EVE Energy
  - **2018-2023**: Total of $1.8 billion to CATL alone
- **Policy Support**:
  - **2009-2017**: Buyer's rebates for EVs
  - **2015-2024**: Subsidies reserved for domestic battery producers
  - **2024-2027**: Extended tax exemption policy for New Energy Vehicles (NEVs)

3. United States

- **Inflation Reduction Act (IRA)**:
  - Extended subsidies to 'friendly' and 'like-minded' countries, including Australia, as part of a strategy to de-risk supply chains from China's dominance.

4. Argentina and Chile

- No specific direct subsidies for lithium mining were identified in the research.
- Both countries have focused more on regulatory frameworks and state participation rather than direct financial subsidies.

Market Context

Price Fluctuations

- **2017-2018**: Lithium prices peaked, leading to increased investment in new projects.
- **2019-2020**: Prices declined significantly, leading to project suspensions and closures, particularly in Australia.
- **2021-2022**: Prices rebounded strongly with the growth of the electric vehicle market.
- **2023-2024**: Prices declined again due to oversupply, leading to production cuts and project delays.

Supply-Demand Balance

- **2023**: The lithium market recorded a surplus of around 175,000 tonnes.
- **2024**: The surplus continued at almost 154,000 tonnes.
- **2024**: Global lithium chemical supply was approximately 9.2% higher than consumption.

Conclusion

The period from 2015 to 2024 saw significant volatility in the lithium market, with prices experiencing both dramatic rises and falls. This volatility led to strategic responses from lithium producers:

1. **Production Adjustments**: While major producers in Chile maintained or expanded production throughout the period, Australian operations were more responsive to price changes, implementing several production cuts and mine closures during downturns, particularly in 2019 and 2024.
2. **Government Intervention**: Governments increasingly recognized lithium as a strategic resource, with Australia and China providing significant subsidies to support their domestic industries. Chile and Argentina focused more on increasing state control and participation rather than direct subsidies.

3. **Geopolitical Dimensions**: The lithium industry became increasingly caught in US-China geopolitical competition, with both powers implementing policies to secure supply chains and reduce dependence on each other.
4. **Environmental and Social Challenges**: Growing concerns about the environmental and social impacts of lithium mining led to increased regulatory scrutiny and legal challenges, particularly in Argentina and Chile.

These trends suggest that the lithium industry is maturing and becoming more strategically important, with production decisions increasingly influenced not just by market forces but also by government policies, geopolitical considerations, and environmental concerns.

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Albemarle's Salar de Atacama Production and Subsidies (2015-2024)

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Albemarle Corporation operates the Salar de Atacama I lithium project in Chile, which has been in operation since before 2015. Key findings about production cuts, closures, and subsidies include:

1. Production Capacity and Expansion:
   - As of 2017-2018, Albemarle could produce approximately 80,000 metric tons of lithium annually from the Salar de Atacama, representing about 36% of global demand in 2017.
   - Albemarle planned to increase its lithium carbonate and lithium chloride capacity in Chile from 44,000 t/yr to 80,000 t/yr of LCE.
   - In May 2024, Chile gave Albemarle an option to boost its lithium quota by 240,000 metric tons.
2. Production Cuts/Adjustments:
   - No specific production cuts at Salar de Atacama were identified between 2015-2024.
   - In January 2024, Albemarle announced workforce reductions and paused expansions amid falling lithium prices, but this was a global strategy rather than specific to Salar de Atacama.
   - In July 2024, Albemarle announced a fresh round of cost cuts due to tumbling lithium prices, but again this was company-wide rather than specific to Salar de Atacama.
   - In July 2024, Albemarle halted expansion of its Kemerton plant in Australia and reduced workforce there by 40%, showing the company's global response to market conditions.
3. Regulatory and Environmental Issues:
   - In 2017, Albemarle expressed concerns about rival SQM's brine extraction levels at Salar de Atacama, suggesting potential impacts on future production.

- Water usage has been a contentious issue, with concerns about the impact of lithium extraction on local water resources.
- Albemarle maintains 150 monitoring wells in the Salar basin, monitored in real time, with representatives from indigenous communities periodically accompanying monitoring activities.

4. Government Payments and Subsidies:
   - In 2022, Albemarle paid the Chilean government over $600 million in taxes and royalties, described by company representatives as "the highest commission in the world to extract lithium."
   - In July 2024, Chilean lithium became eligible for U.S. tax breaks under the Inflation Reduction Act (IRA), providing potential benefits for Albemarle's operations.
   - No direct subsidies to Albemarle's operations in Chile were identified in the research.
5. Government Policy Changes:
   - In April 2023, Chile announced plans to nationalize its lithium industry, but stated it would respect existing contracts. Albemarle's contract runs until 2043.
   - The Chilean government is moving toward a public-private model for lithium extraction, with the state taking a controlling interest in partnerships.
   - Albemarle has expressed openness to partnering with the Chilean state-owned lithium company.
6. Recent Developments:
   - In August 2025, there was an incident at Albemarle's lithium processing plant in Chile (La Negra), which was being investigated, but the plant remained operational.
   - Albemarle continues to operate in Chile with a contract that extends until 2043.

No specific instances of complete closure or idle periods at the Salar de Atacama I project were identified in the research for the 2015-2024 period.

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SQM's Salar de Atacama II Production and Subsidies (2015-2024)

카테고리: production_cuts_and_closures

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SQM (Sociedad Química y Minera de Chile) operates the Salar de Atacama II lithium project in Chile. Key findings about production cuts, closures, and subsidies for the period 2015-2024 include:

1. Production Capacity and Expansion:
   - SQM accounts for approximately 65% of Chile's lithium production.
   - In 2024, SQM planned to expand its production capacity in the Salar de Atacama to 300,000 tonnes per year (tpy) from about 200,000 tpy, despite falling lithium prices.
   - SQM's lithium extraction quota was 349,553 tons plus a remnant of 64,816 tons from the previous quota of 180,001 tons.
   - In March 2024, SQM announced plans to increase lithium production despite low prices in the market.
2. Production Cuts/Workforce Reductions:
   - In June 2025, SQM announced a 5% reduction in its Chilean workforce (approximately 417 positions) due to stagnating lithium prices.
   - The layoffs affected multiple business segments and locations, with approximately 25-30% targeting "general roles" (104-125 positions) and 70-75% affecting supervisory positions (292-313 positions).
   - This workforce reduction was implemented to streamline management layers while maintaining frontline operational capacity.

- SQM reported a drop in net profit in Q4 2023 due to weak lithium prices and expected prices to dip further at the start of 2025.

3. Environmental Commitments:
   - Between 2016 and 2019, SQM reduced its water consumption in the Atacama basin by 25%.
   - The company committed to reducing freshwater consumption by a further 50% by 2030.
   - SQM also committed to reducing brine extractions at Salar de Atacama by 50% by 2030 by optimizing operations.
   - In September 2023, SQM's Salar de Atacama lithium mining operations achieved IRMA 75 certification (Initiative for Responsible Mining Assurance), reportedly becoming the only lithium company to achieve this level worldwide.
4. Government Agreements and Policies:
   - In December 2023, SQM reached a framework agreement to hand over a majority stake in its Chilean brine assets to state-owned Codelco in exchange for extending operations for three more decades.
   - SQM's original contract was set to expire in 2030.
   - In April 2023, Chile announced plans to nationalize its lithium industry, which would eventually transfer control from SQM and Albemarle to a state-owned company.
   - Future lithium contracts would only be issued as public-private partnerships with state control.
   - SQM was reportedly investing $2 billion into sustainable technologies to meet the government's environmental goals.
5. Financial Performance:
   - Miners worldwide, including SQM, have been hit by a nearly 90% drop in lithium prices since their peak in late 2022 due to weaker-than-expected demand for electric vehicles and excess supply.
   - In early 2025, SQM missed profit estimates as lithium prices remained under pressure.
   - For 2025, SQM forecast a significant reduction in potassium production, aligned with its strategy to reduce brine extraction and prioritize high-lithium-content brines.
6. Protests and Social Issues:
   - In January 2024, protests at Chile's lithium salt flats snarled roads to both SQM and Albemarle operations, indicating social tensions related to lithium extraction.

No specific instances of complete closure or idle periods at the Salar de Atacama II project were identified in the research for the 2015-2024 period. The main production adjustment was the workforce reduction announced in June 2025, which was in response to market conditions rather than a direct production cut.

출처: 1. https://www.reuters.com/world/americas/chilean-lithium-miner-sqm-begins-laying-off-5-local-workforce-2025-06-25/ 2. https://www.bloomberg.com/news/articles/2023-12-27/sqm-and-codelco-reach-agreement-for-lithium-mining-in-chile 3. https://www.bloomberg.com/news/articles/2024-03-21/sqm-subira-produccion-de-litio-a-pesar-de-baja-en-los-precios 4. https://www.cnbc.com/2023/05/06/chile-state-led-lithium-policy-central-to-future-of-electric-vehicles.html 5. https://www.cnbc.com/2023/04/21/chile-plans-to-nationalize-its-vast-lithium-industry.html 6. https://sqmlitio.com/wp-content/uploads/2021/05/SQM-Sustainable-Lithium-English-20210504.pdf 7. https://pubs.rsc.org/en/content/articlehtml/2024/ma/d4ma00625a 8. https://discoveryalert.com.au/news/sqms-workforce-reduction-lithium-market-2025/ 9. https://www.mining.com/web/sqm-misses-profit-estimates-as-lithium-prices-remain-under-pressure/

Olaroz Lithium Project Production and Subsidies (2015-2024)

카테고리: production_cuts_and_closures

수집 시간: 2025-08-18 10:37:16

No topic

카테고리: No category

수집 시간: No time information

The Olaroz lithium project in Argentina is operated by Sales de Jujuy SA. Key findings about production cuts, closures, and subsidies for the period 2015-2024 include:

1. Project Establishment and Production:
   - The Sales de Jujuy project in the Olaroz salt flat in Jujuy Province, Argentina, began operations in 2015.
   - This project marked a turning point for Argentina's lithium industry, making the country the most dynamic lithium-producing country in the world at that time.
   - In 2016, the Olaroz project represented 6% of global lithium production.

- According to national statistics, total production in 2016 was 30,335 tonnes LCE (lithium carbonate equivalent), of which 11,845 tonnes were generated by Sales de Jujuy's project.
- By the end of 2017, the project was expected to reach its full production capacity of 17,500 tonnes per year.

2. Ownership Structure:
   - Sales de Jujuy is a joint venture company with multiple stakeholders.
   - The project is located in the Olaroz-Caucharí salt flat in the department of Susques, Jujuy Province.
3. Production Growth in Argentina:
   - Argentina's lithium mining industry showed an impressive increase in production of over 100%, from 8,630 MT in 2023 to 18,000 MT in 2024.
   - The country's lithium reserves increased from 310,000 MT in 2023 to an estimated 480,000 MT in 2024, according to the U.S. Geological Survey.
   - Experts anticipated that Argentina had the potential to increase its lithium production tenfold in just a few years.
4. Industry Challenges:
   - In October 2024, Reuters reported that Argentina's lithium hunters were scaling back operations as the electric vehicle shift slowed.
   - Firms across Argentina cut staff, slashed spending, and halted exploration projects due to plunging lithium prices.
   - The value of lithium assets fell, leaving some firms vulnerable to takeover.
   - However, no specific production cuts or closures were reported for the Olaroz project in this period.
5. COVID-19 Impact:
   - Argentina's fledgling lithium sector was affected by the coronavirus pandemic, though specific impacts on the Olaroz project were not detailed in the available sources.
6. Community Relations:
   - The project has engaged with local indigenous communities in the department of Susques.
   - Research has been conducted to investigate the local impacts of lithium projects in the Argentine Puna, including Sales de Jujuy's operations.
7. Government Policies:
   - Argentina's approach to lithium mining differs from neighboring Chile and Bolivia.
   - While Chile has moved toward state control and Bolivia has full state control of the sector, Argentina has generally granted concessions for companies to operate.
   - No specific subsidies for the Olaroz project were identified in the research.

No specific instances of production cuts, complete closure, or idle periods at the Olaroz project were identified in the research for the 2015-2024 period. While the broader Argentine lithium industry faced challenges in 2024 due to falling prices, with some companies reducing operations, the available sources did not specify whether Sales de Jujuy's Olaroz project implemented any production cuts during this period.

출처: 1. https://farn.org.ar/wp-content/uploads/2019/05/DOC_LITHIUM_ENGLISH-1.pdf 2. https://goodelectronics.org/wp-content/uploads/sites/3/2019/05/DOC_LITHIUM_ENGLISH.pdf 3. https://erlacs.org/articles/11121/files/6762c08c5bf28.pdf 4. https://www.developmentaid.org/news-stream/post/170661/five-major-lithium-producing-countries 5. https://www.reute lithium-hunters-scale-back-ev-shift-slows-2024-10-24/

Salar del Hombre Muerto Lithium Project Production and Subsidies (2015-2024)

카테고리: production_cuts_and_closures

수집 시간: 2025-08-18 10:37:54

No topic

카테고리: No category

수집 시간: No time information

The Salar del Hombre Muerto lithium project in Argentina is operated by Livent (formerly FMC Corporation). Key findings about production cuts, closures, and subsidies for the period 2015-2024 include:

1. Project History and Operations:
   - Livent has been extracting lithium brine at Salar del Hombre Muerto in Argentina for more than 25 years (since approximately 1997).
   - The project is located in Catamarca Province, Argentina.
   - It is considered one of the lowest-cost resources globally for lithium carbonate and lithium chloride.

- As of 2023, Livent had a current capacity of approximately 20,000 metric tons of lithium carbonate and 9,000 metric tons of lithium chloride at this site.
- Livent holds mineral concession rights through its subsidiary MdA to extract lithium brine without time limitations.

2. Production Facilities:
   - Livent operates two main facilities in Argentina:
     a) Fénix facility: Located at the Salar del Hombre Muerto in Catamarca province, producing lithium carbonate from high-quality lithium brine utilizing a proprietary selective adsorption direct lithium extraction (DLE) technology process since 1997.
     b) Güemes facility: Located in Salta, Argentina, producing lithium chloride from high-quality lithium brine from the Salar del Hombre Muerto. The lithium chloride feeds the production of lithium metal used as feedstock by Livent's network of butyllithium manufacturing facilities.
3. Expansion Projects:
   - In May 2024, the first phase of a 10,000 metric ton expansion at the Fénix facility was completed.
   - Livent has been working on expansion projects at Salar del Hombre Muerto, though specific details on all phases were not provided in the sources.
4. Corporate Changes:
   - In January 2024, Livent merged with Australian company Allkem to form a new company called Arcadium Lithium, which became the third-largest lithium producer in the world.
   - Following the merger, the Salar del Hombre Muerto operations continued under the Arcadium Lithium umbrella.
5. Environmental and Legal Issues:
   - In March 2024, an Argentine court in the key lithium region halted new permits over environmental concerns. The ruling involved the Los Patos River-Salar del Hombre Muerto area, where Arcadium Lithium (formerly Livent) has its project.
   - The court decision followed tensions over water resources in the region.
   - Livent has faced long-running conflict with communities living near the salt flats they control, who allege that the company's activities are causing water stress.
   - According to researcher Melisa Argento from the National Council for Scientific and Technical Research (CONICET), "Livent has been extracting lithium from the Salar del Hombre Muerto for 25 years… Communities have been complaining for a long time that the plains of the nearby Trapiche River are dry."
6. Market Challenges:
   - In October 2024, Reuters reported that Argentina's lithium hunters were scaling back operations as the electric vehicle shift slowed.
   - Arcadium CEO Paul Graves told analysts: "We must adapt to the realities of the market we find ourselves in today and the pace at which we can responsibly invest capital," when announcing cuts.
   - However, specific production cuts at the Salar del Hombre Muerto operation were not detailed in the available sources.
7. Resource Potential:
   - According to development aid sources, Argentina's Salar del Hombre Muerto area has huge lithium brine deposits that will last for a minimum of 75 years.
   - As of December 31, 2022, Livent reported the following resource estimates for Salar del Hombre Muerto:
     - Measured & Indicated: 1,328 thousand metric tons of lithium (7,071 thousand metric tons LCE)
     - Inferred: 892 thousand metric tons of lithium (4,749 thousand metric tons LCE)
     - Total: 2,220 thousand metric tons of lithium (11,820 thousand metric tons LCE)
8. Government Subsidies:
   - No specific subsidies for the Salar del Hombre Muerto project were identified in the research.
   - Argentina's approach to lithium mining has generally been to grant concessions for companies to operate, rather than providing direct subsidies.

No specific instances of production cuts, complete closure, or idle periods at the Salar del Hombre Muerto project were identified in the research for the 2015-2024 period. While the broader Argentine lithium industry faced challenges in 2024 due to falling prices, with some companies reducing operations, the available sources did not specify whether Livent/Arcadium implemented any production cuts at this specific project during this period.

출처: 1. https://www.reuters.com/world/americas/argentine-court-key-lithium-region-halts-new-permits-over-environmental-concerns-2024-03-14/ 2. https://www.reuters.com/markets/commodities/argentinas-lithium-hunters-scale-back-ev-shift-slows-2024-10-24/ 3. https://s201.q4cdn.com/709125885/files/doc_presentation/Livent/2023/Livent-Investor-Presentation-v2023-10-1.pdf 4. https://arcadiumlithium.com/wp-content/uploads/2024/09/2023-Arcadium-Lithium-Sustainability-Report.pdf 5. https://dialogue.earth/en/business/392580-who-controls-argentinas-lithium/ 6. https://www.developmentaid.org/news-stream/post/170661/five-major-lithium-producing-countries

Cauchari-Olaroz Lithium Project Production and Subsidies (2015-2024)

카테고리: production_cuts_and_closures

수집 시간: 2025-08-18 10:38:29

No topic

카테고리: No category

수집 시간: No time information

The Cauchari-Olaroz lithium project in Argentina is operated by Minera Exar. Key findings about production cuts, closures, and subsidies for the period 2015-2024 include:

1. Project Development and Commissioning:
   - The Cauchari-Olaroz project is located in Jujuy Province, Argentina, in the Lithium Triangle region.
   - The project was commissioned in 2024, making it a relatively new operation compared to other established lithium projects.
   - It has a nameplate capacity of 40,000 tonnes per annum (tpa) of battery-grade lithium carbonate.
   - The project has significant potential for expansion beyond its initial capacity.
2. Ownership Structure:
   - The project is owned by a joint venture consisting of:
     - Ganfeng Lithium (46.7%)
     - Lithium Argentina (44.8%)
     - JEMSE (Jujuy Energía y Minería Sociedad del Estado, a Jujuy provincial government company) (8.5%)
   - The holding company for the operation is Minera Exar.
   - In April 2021, JEMSE was incorporated into Minera Exar's project, making use of its 8.5% participation option.
   - Ganfeng Lithium Group, headquartered in Jiangxi, China, owns 46.66% of Cauchari-Olaroz, which renders any lithium mined there ineligible for U.S. Inflation Reduction Act (IRA) benefits.
3. Project Status and Technology:
   - The project utilizes brine evaporation ponds and processing facilities.
   - It employs innovative technologies like Direct Lithium Extraction (DLE).
   - The project prioritizes efficient operations, water conservation, ecosystem preservation, and strong partnerships with local communities.
   - Agreements are in place for 80% of offtake at market prices.
4. Market Position:
   - Cauchari-Olaroz is Argentina's biggest operating lithium mine.
   - It is part of Argentina's plan to significantly increase lithium production capacity.
   - According to Bloomberg, Argentina was expected to overtake Chile in lithium production around 2027, with Lithium Americas Corp. and Ganfeng Lithium Co.'s Cauchari-Olaroz likely to be a significant contributor to this growth.
5. Industry Challenges:
   - In October 2024, Reuters reported that Argentina's lithium hunters were scaling back operations as the electric vehicle shift slowed.
   - Firms across Argentina cut staff, slashed spending, and halted exploration projects due to plunging lithium prices.
   - The value of lithium assets fell, leaving some firms vulnerable to takeover.
   - However, no specific production cuts or closures were reported for the Cauchari-Olaroz project in this period.
6. Production Expansion in Argentina:
   - In June 2024, Bloomberg reported that Argentina was about to "unleash a wave of lithium" with four new projects beginning to produce lithium in the weeks and months ahead, which would almost double production capacity in the country.
   - This expansion was occurring despite a global glut in lithium supply.
   - The report did not specify if Cauchari-Olaroz was one of these four projects or if it was already counted in the existing capacity.
7. Community and Environmental Aspects:
   - Research has been conducted to investigate the local impacts of lithium projects in the Argentine Puna, including Minera Exar's operations.
   - The project is located in an area where indigenous communities of the department of Susques largely cooperate with the mining companies.

- In June 2023, while protests against the state escalated in neighboring communities of the Salinas Grandes-Laguna Guayatayoc, the Cauchari-Olaroz project continued operations.

8. Government Policies and Subsidies:
   - No specific subsidies for the Cauchari-Olaroz project were identified in the research.
   - Argentina's approach to lithium mining differs from neighboring Chile and Bolivia, with Argentina generally granting concessions for companies to operate rather than providing direct subsidies.
   - The provincial government of Jujuy has taken a more active role in managing lithium mining compared to other Argentine provinces like Salta and Catamarca, which foster more neoliberal mining policies.

No specific instances of production cuts, complete closure, or idle periods at the Cauchari-Olaroz project were identified in the research for the 2015-2024 period. This is likely because the project was only commissioned in 2024 and had not been operating long enough to experience such events. While the broader Argentine lithium industry faced challenges in 2024 due to falling prices, with some companies reducing operations, the available sources did not specify whether Minera Exar implemented any production cuts at this specific project during this period.

출처: 1. https://www.reuters.com/markets/commodities/argentinas-lithium-hunters-scale-back-ev-shift-slows-2024-10-24/ 2. https://www.bloomberg.com/news/articles/2022-05-24/lithium-demand-for-ev-batteries-pushes-argentina-mining 3. https://www.bloomberg.com/news/articles/2024-06-28/argentina-is-about-to-unleash-a-wave-of-lithium-in-a-global-glut 4. https://www.lithium-argentina.com/projects/cauchari-olaroz 5. https://farn.org.ar/wp-content/uploads/2019/05/DOC_LITHIUM_ENGLISH-1.pdf 6. https://erlacs.org/articles/11121/files/6762c08c5bf28.pdf 7. https://www.csis.org/analysis/leveraging-argentinas-mineral-resources-economic-growth

Chaerhan Lake Lithium Project Production and Subsidies (2015-2024)

카테고리: production_cuts_and_closures

수집 시간: 2025-08-18 10:39:04

No topic

카테고리: No category

수집 시간: No time information

The Chaerhan Lake (also spelled Qarhan) lithium project in China is operated by Qinghai Salt Lake Industry Co., Ltd. Key findings about production cuts, closures, and subsidies for the period 2015-2024 include:

1. Project Location and Operators:
   - Chaerhan Salt Lake is situated in Golmud City, Qinghai Province, China.
   - The lake is operated by two major entities:
     a) Qinghai Salt Lake Industry Co., Ltd.
     b) Tibet Mining Co., Ltd.
   - Both companies contribute substantially to the region's economic development through lithium carbonate extraction.
2. Production Capacity and Growth:
   - According to Mining Technology, the Chaerhan Lake Mine produced an estimated 27.5 thousand tonnes of lithium in 2023.
   - Tibet Mining Co., Ltd. reported an annual production capacity of 12,000 tons of lithium carbonate equivalent (LCE) in 2023.
   - Qinghai Salt Lake Industry Co., Ltd. was operating with a capacity of 30,000 tons of LCE per annum as of 2023-2024.
   - The company was expanding its production capabilities, with an additional 40,000 tons per year expected to come online by the end of 2024.
   - The total production from Chaerhan Salt Lake was on an upward trajectory, with an estimated output of 52,000 tons of LCE in 2024.
   - Projections for 2025 indicated a significant increase to 82,000 tons of LCE.
3. Corporate Changes:
   - China Minmetals acquired Qinghai Salt Lake Industry's Qarhan II (part of the Chaerhan Lake operations) in 2024, which was described as an example of industry improvements focusing investment on higher quality resources.
4. Production Cuts and Closures:
   - In July 2025, Zangge Mining's Qarhan I (part of the Chaerhan Lake operations) was ordered to halt its 10,000 tons/year lithium carbonate production capacity by the local administration.

    - The company was required to complete an application for lithium extraction, and only after all mining permits were fully compliant could it apply to the Haixi Prefecture Salt Lake Administration for resumption of production.
    - This closure was part of China's broader efforts to reform its domestic lithium supply and address overcapacity issues.
5. Industry-wide Production Adjustments:
    - In September 2024, CLPC China Lithium Products Technology Co., Ltd. issued a notice that its holding subsidiary company would carry out a 15-day production suspension and maintenance work, while its wholly-owned subsidiary would carry out a 10-day production suspension and maintenance work.
    - This was part of a broader trend in China's lithium industry, where companies were implementing production cuts to shore up market sentiment amid falling prices.
    - According to CRU Group analysis, Jiangxi's mines that did not have permits to extract lithium, nor were licensed by the Ministry of Natural Resources (MNR), were deemed most at risk of closure.
    - Besides these operations, two other brine producers fell into this category - Zangge's Qarhan I and Qinghai Hengxinrong's West Taijnar II.
    - Together, these operations were forecast to account for approximately 175,000 tons LCE of supply in 2025, before Qarhan I's closure.
6. Market Conditions and Government Policy:
    - The slowdown in the growth trend of lithium battery industry demand in 2024 was considered a high probability event.
    - Against the backdrop of excess capacity in the entire market, the clearance of some high-cost mining capacity was seen as the only way for prices to bottom out.
    - However, as of 2024, large-scale production cuts and suspensions had not yet occurred across the industry.
    - China's domestic salt lake production was expected to increase to 135,000 tons of LCE in 2024, a year-on-year increase of 18.4%, according to statistics from Antai Technology.
7. Government Subsidies:
    - No specific subsidies for the Chaerhan Lake project were identified in the research.
    - However, China's lithium industry has generally benefited from government support and policies promoting the development of the electric vehicle and battery sectors.

The most significant production cut identified for the Chaerhan Lake operations was the July 2025 halt of Zangge Mining's Qarhan I facility (10,000 tons/year capacity), which was ordered by local authorities due to compliance issues with mining permits. This appears to be part of China's broader effort to reform its domestic lithium supply chain and address overcapacity issues in the industry.

출처: 1. https://www.linkedin.com/pulse/global-brine-lithium-production-capacity-costs-2024-2025-xuan-cewang-lwbac 2. https://www.mining-technology.com/marketdata/ten-largest-lithiums-mines/ 3. http://lithium-chemical.com/Company/458.html 4. https://www.crugroup.com/en/communities/thought-leadership/2025/China-has-domestic-lithium-supply-reform-squarely-in-its-sights/

Yiliping Lithium Project Production and Subsidies (2015-2024)

카테고리: production_cuts_and_closures

수집 시간: 2025-08-18 10:39:43

No topic

카테고리: No category

수집 시간: No time information

The Yiliping lithium project in China is operated by China Minmetals Corp. While specific information about the Yiliping project was limited in the sources, the research revealed important context about China's lithium industry and production adjustments during the 2015-2024 period:

1. Project Information:
    - Yiliping is a lithium project in China operated by China Minmetals Corp.
    - Specific details about the project's capacity, production history, and exact location were not provided in the available sources.
2. Chinese Lithium Industry Context:

- China has secured an 80% share of global lithium chemical production, 78% of cathode production, and 70% of cell manufacturing for the electric car industry, despite holding less than 7% of the world's lithium reserves.
- In 2022, China held approximately 75% of global battery manufacturing capacity.
- The lithium sector has been grappling with weaker-than-anticipated growth in demand for electric vehicles and an oversupplied market.
- After soaring to record highs in 2022, prices of lithium tumbled almost 90% by 2024-2025, prompting producers around the world to scale back investment plans.

3. Production Cuts and Mine Closures in China:
   - While specific information about production cuts at Yiliping was not found, there were several notable production cuts and mine closures in China's lithium industry during this period:
     a) In August 2025, CATL suspended production at its Jianxiawo mine in Yichun, Jiangxi province, after its license expired. This mine accounted for about 3-6% of global lithium output.
     b) In July 2025, Zangge Mining halted lithium production at a mine in Qinghai province following orders from local officials.
     c) In September 2024, CLPC China Lithium Products Technology Co. announced a 15-day production suspension and maintenance work at its mining subsidiary and a 10-day suspension at its mineral processing plant.
4. Market Conditions and Industry Response:
   - A nearly 86% plunge in prices of lithium over 2023-2024 forced companies to mothball mines across the world.
   - By January 2025, analysts expected to see a price recovery for lithium as the curtailments seen in 2024, and the possibility of further curtailments, would significantly reduce the market surplus.
   - The price collapse affected Chinese companies as well, forcing battery giant CATL to suspend production at certain mines.
   - In October 2024, a U.S. official accused China of oversupplying lithium to eliminate rivals, suggesting that China was engaging in "predatory pricing" in the lithium market.
5. Government Policies and Reforms:
   - By mid-2025, China had toughened its rhetoric against overcapacity, sparking hope among investors that supply-side reforms could rein in output of lithium and other commodities.
   - These policy shifts led to production suspensions at several mines, including Zangge Mining's operation in Qinghai.
   - The Chinese government had previously introduced subsidy-centric policies that reserved benefits for domestic battery producers, which marked a turning point for the battery industry in China.
   - These policies introduced fresh funds for mining, resources exploration, and processing machinery.
6. EV Subsidies and Market Impact:
   - China doubled EV subsidies in July 2024, which contributed to a lithium price rally late that year.
   - By December 2024, more than 5 million cars sold in China had benefited from these incentives.
   - Analysts and traders expected these subsidies to continue supporting lithium prices in 2025.
   - A buyer at a mid-sized cathode material plant in China stated: "The uptick in lithium trade business in the fourth quarter of 2024 can be undeniably attributed to the policy of providing subsidies."
7. Yichun Mining Hub:
   - While not specifically about Yiliping, the sources mentioned that Yichun in central China had become a major lithium mining hub.
   - The growl of extractors echoed across the vast mountaintop mines of Yichun, with a procession of heavy-duty trucks rolling down the valley to waiting refineries.
   - CATL's closure of its mine in Yichun was the first publicly announced in this important lithium mining region.

No specific information about production cuts, closures, or subsidies directly related to the Yiliping project operated by China Minmetals Corp was found in the available sources. However, the broader context of China's lithium industry shows that many producers were affected by falling prices and government policies aimed at addressing overcapacity during the 2023-2025 period. Without specific data on Yiliping, it cannot be determined whether this particular project implemented production cuts or received subsidies during the 2015-2024 timeframe.

출처: 1. https://www.reuters.com/markets/commodities/lithium-prices-stabilise-2025-mine-closures-china-ev-sales-ease-glut-analysts-2025-01-13/ 2. https://www.reuters.com/business/energy/chinas-catl-suspends-operations-major-lithium-mine-seeks-new-license-2025-08-11/ 3. https://www.reuters.com/markets/commodities/china-is-oversupplying-lithium-eliminate-rivals-us-official-says-2024-10-08/ 4. https://www.bloomberg.com/news/articles/2025-08-11/lithium-stocks-surge-as-chinese-mine-closes-after-permit-expires 5. https://www.bloomberg.com/news/articles/2025-05-22/us-china-trade-war-beijing-bets-on-homegrown-lithium-for-high-tech-economy 6. https://www.spglobal.com/commo
insights/en/news-research/latest-news/metals/092324-factbox-chinas-lithium-industry-eyes-output-cuts-to-shore-up-

market-sentiment 7. https://orcasia.org/article/602/chinas-monopoly-over-lithiums-upstream-and-downstream-supply-chain

East Taijnar Lithium Project Production and Subsidies (2015-2024)

카테고리: production_cuts_and_closures

수집 시간: 2025-08-18 10:40:08

No topic

카테고리: No category

수집 시간: No time information

The East Taijnar lithium project in China is operated by Qinghai Dongtai Jinel Lithium Resources Co., Ltd. Information about this specific project was very limited in the available sources, but some context about China's lithium industry and related projects was found:

1. Project Information:
   - East Taijnar is a lithium project in China operated by Qinghai Dongtai Jinel Lithium Resources Co., Ltd.
   - Specific details about the project's capacity, production history, and exact location were not provided in the available sources.
2. Related Projects and Industry Context:
   - While East Taijnar itself was not specifically mentioned in the available sources regarding production cuts or subsidies, a related project called West Taijnar II was mentioned.
   - According to CRU Group analysis from 2025, Qinghai Hengxinrong's West Taijnar II was identified as one of the brine producers that did not have a permit to extract lithium, nor was it licensed by the Ministry of Natural Resources (MNR).
   - West Taijnar II was described as producing chemicals using third-party feedstock.
   - Operations like West Taijnar II were deemed at risk of closure as part of China's domestic lithium supply reform efforts.
3. Chinese Lithium Industry Reforms:
   - In 2025, China was implementing reforms in its domestic lithium supply chain.
   - These reforms were expected to enable gradual industry improvements, such as focusing investment on higher quality resources.
   - Jiangxi's mines that did not have permits to extract lithium or were not licensed by the MNR were deemed most at risk of closure.
   - Besides these operations, two other brine producers fell into this category - Zangge's Qarhan I and Qinghai Hengxinrong's West Taijnar II.
   - Together, these operations were forecast to account for approximately 175,000 tons LCE of supply in 2025, before Qarhan I's closure.
4. Notable Production Cuts in the Industry:
   - In July 2025, Zangge Mining's Qarhan I was ordered to halt its 10,000 tons/year lithium carbonate production capacity by the local administration.
   - The company was required to complete an application for lithium extraction, and only after all mining permits were fully compliant could it apply to the Haixi Prefecture Salt Lake Administration for resumption of production.

No specific information about production cuts, closures, or subsidies directly related to the East Taijnar project operated by Qinghai Dongtai Jinel Lithium Resources Co., Ltd. was found in the available sources. The limited information available only mentioned a related project (West Taijnar II) as being at risk of closure due to permit issues. Without specific data on East Taijnar, it cannot be determined whether this particular project implemented production cuts or received subsidies during the 2015-2024 timeframe.

출처: 1. https://www.crugroup.com/en/communities/thought-leadership/2025/China-has-domestic-lithium-supply-reform-squarely-in-its-sights/

West Taijnar I Lithium Project Production and Subsidies (2015-2024)

카테고리: production_cuts_and_closures

수집 시간: 2025-08-18 10:40:29

No topic

카테고리: No category

수집 시간: No time information

The West Taijnar I lithium project in China is operated by Qinghai CITIC Guoan Technology Development Co., Ltd. Information about this specific project was very limited in the available sources, but some context about China's lithium industry and related projects was found:

1. Project Information:
   - West Taijnar I is a lithium project in China operated by Qinghai CITIC Guoan Technology Development Co., Ltd.
   - Specific details about the project's capacity, production history, and exact location were not provided in the available sources.
2. Related Projects and Industry Context:
   - While West Taijnar I itself was not specifically mentioned in the available sources regarding production cuts or subsidies, a related project called West Taijnar II was mentioned.
   - According to CRU Group analysis from 2025, Qinghai Hengxinrong's West Taijnar II was identified as one of the brine producers that did not have a permit to extract lithium, nor was it licensed by the Ministry of Natural Resources (MNR).
   - West Taijnar II was described as producing chemicals using third-party feedstock.
   - Operations like West Taijnar II were deemed at risk of closure as part of China's domestic lithium supply reform efforts.
3. Chinese Lithium Industry Reforms:
   - In 2025, China was implementing reforms in its domestic lithium supply chain.
   - These reforms were expected to enable gradual industry improvements, such as focusing investment on higher quality resources.
   - Jiangxi's mines that did not have permits to extract lithium or were not licensed by the MNR were deemed most at risk of closure.
   - Besides these operations, two other brine producers fell into this category - Zangge's Qarhan I and Qinghai Hengxinrong's West Taijnar II.
   - Together, these operations were forecast to account for approximately 175,000 tons LCE of supply in 2025, before Qarhan I's closure.
4. Notable Production Cuts in the Industry:
   - In July 2025, Zangge Mining's Qarhan I was ordered to halt its 10,000 tons/year lithium carbonate production capacity by the local administration.
   - The company was required to complete an application for lithium extraction, and only after all mining permits were fully compliant could it apply to the Haixi Prefecture Salt Lake Administration for resumption of production.

No specific information about production cuts, closures, or subsidies directly related to the West Taijnar I project operated by Qinghai CITIC Guoan Technology Development Co., Ltd. was found in the available sources. The limited information available only mentioned a related project (West Taijnar II) as being at risk of closure due to permit issues. Without specific data on West Taijnar I, it cannot be determined whether this particular project implemented production cuts or received subsidies during the 2015-2024 timeframe.

출처: 1. https://www.crugroup.com/en/communities/thought-leadership/2025/China-has-domestic-lithium-supply-reform-squarely-in-its-sights/

West Taijnar II Lithium Project Production and Subsidies (2015-2024)

카테고리: production_cuts_and_closures

수집 시간: 2025-08-18 10:40:50

No topic

카테고리: No category

수집 시간: No time information

The West Taijnar II lithium project in China is operated by Qinghai Hengxinrong. Information about this project was limited in the available sources, but some specific details were found:

1. Project Information:
   - West Taijnar II is a lithium project in China operated by Qinghai Hengxinrong.
   - The project produces chemicals using third-party feedstock.
   - Specific details about the project's capacity and production history were not provided in the available sources.
2. Regulatory Status and Risks:
   - According to CRU Group analysis from 2025, Qinghai Hengxinrong's West Taijnar II was identified as one of the brine producers that did not have a permit to extract lithium.
   - The project was not licensed by the Ministry of Natural Resources (MNR).
   - Due to these regulatory issues, West Taijnar II was deemed at risk of closure as part of China's domestic lithium supply reform efforts.
3. Industry Context and Supply Impact:
   - West Taijnar II was one of several operations identified as being at risk due to permit issues.
   - Jiangxi's mines that did not have permits to extract lithium or were not licensed by the MNR were deemed most at risk of closure.
   - Besides these operations, another brine producer fell into this category - Zangge's Qarhan I.
   - Together, these operations (including West Taijnar II) were forecast to account for approximately 175,000 tons LCE of supply in 2025, before Qarhan I's closure.
4. Chinese Lithium Industry Reforms:
   - In 2025, China was implementing reforms in its domestic lithium supply chain.
   - These reforms were expected to enable gradual industry improvements, such as focusing investment on higher quality resources.
   - As part of these reforms, operations without proper permits were being scrutinized.
5. Notable Production Cuts in the Industry:
   - In July 2025, Zangge Mining's Qarhan I (another operation in a similar regulatory situation to West Taijnar II) was ordered to halt its 10,000 tons/year lithium carbonate production capacity by the local administration.
   - The company was required to complete an application for lithium extraction, and only after all mining permits were fully compliant could it apply to the Haixi Prefecture Salt Lake Administration for resumption of production.

While West Taijnar II was identified as being at risk of closure due to permit issues, the available sources did not confirm whether any actual production cuts or closures had been implemented at this project during the 2015-2024 period. The information suggests that as of 2025, the project was still operating but faced regulatory risks that could lead to future closure or production adjustments. No information about subsidies specifically related to West Taijnar II was found in the available sources.

출처: 1. https://www.crugroup.com/en/communities/thought-leadership/2025/China-has-domestic-lithium-supply-reform-squarely-in-its-sights/

Summary of Lithium Projects Production Cuts, Closures, and Subsidies (2015-2024)

카테고리: summary

수집 시간: 2025-08-18 10:41:36

No topic

카테고리: No category

수집 시간: No time information

Summary of Lithium Projects Production Cuts, Closures, and Subsidies (2015-2024)

Based on the research conducted on ten high-priority lithium projects across South America and Asia, the following key findings have emerged regarding production cuts, closures/idle periods, and subsidies between 2015 and 2024:

1. Production Cuts and Workforce Reductions

South American Projects:

- **SQM (Salar de Atacama II, Chile)**: In June 2025, SQM announced a 5% reduction in its Chilean workforce (approximately 417 positions) due to stagnating lithium prices. This affected multiple business segments, with 25-30% targeting "general roles" and 70-75% affecting supervisory positions.
- **Albemarle (Salar de Atacama I, Chile)**: No specific production cuts at Salar de Atacama were identified between 2015-2024. However, in January 2024, Albemarle announced global workforce reductions and paused expansions amid falling lithium prices, and in July 2024, implemented further cost cuts due to tumbling lithium prices.
- **Argentine Projects**: In October 2024, Reuters reported that Argentina's lithium hunters were scaling back operations as the electric vehicle shift slowed. Firms cut staff, slashed spending, and halted exploration projects due to plunging lithium prices. However, no specific production cuts were reported for Olaroz, Salar del Hombre Muerto, or Cauchari-Olaroz projects.

Asian Projects:

- **Zangge Mining's Qarhan I (Chaerhan Lake, China)**: In July 2025, this operation was ordered to halt its 10,000 tons/year lithium carbonate production capacity by local authorities due to permit issues.
- **CATL's Jianxiawo Mine (China)**: In August 2025, CATL suspended production at its mine in Yichun, Jiangxi province, after its license expired. This mine accounted for about 3-6% of global lithium output.
- **West Taijnar II (China)**: This project operated by Qinghai Hengxinrong was identified as being at risk of closure due to not having proper permits to extract lithium, though no confirmed closure was reported by 2024.

2. Closures and Idle Periods

No complete closures or extended idle periods were identified for the major South American projects (Salar de Atacama I & II, Olaroz, Salar del Hombre Muerto, and Cauchari-Olaroz) during the 2015-2024 period.

In China, the only confirmed closure was Zangge Mining's Qarhan I in July 2025, which was ordered to halt operations until it completed proper licensing procedures. CATL's mine closure in August 2025 was described as temporary while seeking license renewal.

3. Subsidies and Government Support

South American Projects:

- **Chile**: No direct subsidies to lithium operations were identified. Instead, companies like Albemarle and SQM paid significant amounts to the Chilean government. In 2022, Albemarle paid over $600 million in taxes and royalties to the Chilean government, described as "the highest commission in the world to extract lithium."
- **Argentina**: No specific subsidies for Argentine lithium projects were identified. Argentina's approach to lithium mining has generally been to grant concessions for companies to operate rather than providing direct subsidies.

Asian Projects:

- **China**: While specific subsidies for individual projects were not identified, China's lithium industry has benefited from government policies supporting the electric vehicle and battery sectors. In July 2024, China doubled EV subsidies, which contributed to a lithium price rally late that year. By December 2024, more than 5 million cars sold had benefited from these incentives.

4. Government Policy Impacts

Chile:

- In April 2023, Chile announced plans to nationalize its lithium industry but stated it would respect existing contracts (Albemarle's runs until 2043, SQM's until 2030).
- The government is moving toward a public-private model for lithium extraction, with the state taking a controlling interest in partnerships.

- In December 2023, SQM reached a framework agreement to hand over a majority stake in its Chilean brine assets to state-owned Codelco in exchange for extending operations for three more decades.

Argentina:

- Argentina's approach to lithium mining differs from Chile and Bolivia, with Argentina generally granting concessions for companies to operate.
- In March 2024, an Argentine court in the key lithium region halted new permits over environmental concerns, affecting the Los Patos River-Salar del Hombre Muerto area.

China:

- By mid-2025, China had toughened its rhetoric against overcapacity, leading to production suspensions at several mines.
- China's domestic lithium supply reforms were targeting operations without proper permits, with Jiangxi's mines and certain brine producers deemed most at risk.

5. Market Conditions

The global lithium market experienced a dramatic price collapse, with prices falling nearly 90% from their peak in late 2022 to 2024-2025. This price decline was driven by: - Chinese oversupply - Weaker-than-expected demand for electric vehicles - Excess production capacity

This market downturn led to companies worldwide scaling back investment plans, reducing workforces, and in some cases, mothballing mines. By January 2025, analysts expected to see a price recovery as production curtailments reduced the market surplus.

Conclusion

Between 2015 and 2024, major lithium projects in South America showed remarkable resilience, with no significant production cuts or closures despite challenging market conditions. The adjustments were primarily workforce reductions and paused expansions rather than actual production cuts.

In contrast, Chinese lithium operations faced more direct interventions, with government-ordered closures for projects lacking proper permits, particularly in 2025 as part of China's domestic lithium supply reforms.

Regarding subsidies, the research found that South American projects operated without direct government subsidies, while China's lithium industry benefited from broader government support for the electric vehicle sector, particularly through consumer subsidies that indirectly supported demand for lithium products.

출처: 1. https://www.reuters.com/markets/commodities/tumbling-lithium-prices-push-albemarle-fresh-round-cost-cuts-2024-07-31/ 2. https://www.reuters.com/world/americas/chilean-lithium-miner-sqm-begins-laying-off-5-local-workforce-2025-06-25/ 3. https://www.reuters.com/markets/commodities/argentinas-lithium-hunters-scale-back-ev-shift-slows-2024-10-24/ 4. https://www.reuters.com/business/energy/chinas-catl-suspends-operations-major-lithium-mine-seeks-new-license-2025-08-11/ 5. https://www.crugroup.com/en/communities/thought-leadership/2025/China-has-domestic-lithium-supply-reform-squarely-in-its-sights/ 6. https://www.bloomberg.com/news/artic 08-11/lithium-stocks-surge-as-chinese-mine-closes-after-permit-expires 7. https://www.cnbc.com/2023/05/06/chile-state-led-lithium-policy-central-to-future-of-electric-vehicles.html

No topic

카테고리: No category

수집 시간: No time information

No topic

카테고리: No category

수집 시간: No time information

상세 분석 결과

Lithium Projects Data Extraction and Analysis

실행 시간: 2025-08-07 16:03:55

Extracted and analyzed data from data/LithiumProduction/Project_list_250611.xlsx Total projects identified: 74 Projects by region: Africa: 10, Asia: 32, Europe: 1, North America: 6, Oceania: 10, South America: 15 Projects by status: Dev: 6, Ops: 68 Top countries: China: 31, Australia: 10, Argentina: 8, Zimbabwe: 6, Brazil: 5

Lithium Projects Search Strategy Development

실행 시간: 2025-08-07 16:04:58

Created search strategy and research templates for investigating lithium projects. The search strategy includes: - Keyword combinations for production cuts, closures, and subsidies - Information sources including news media, research institutions, and company reports - Data collection format for each project - Regional breakdown of projects: Asia (32), South America (15), Oceania (10), North America (6), Africa (10), Europe (1)

Lithium Projects Visualization

실행 시간: 2025-08-07 16:05:35

Created visualizations for lithium project data analysis: - Regional distribution of projects: Asia (32), South America (15), etc. - Country distribution: China (32) has the most projects - Project status: 68 operational projects, 6 in development - Start year distribution: Projects started from 2015 to 2025 - Extraction types: Evaporation, Direct Extraction, Evaporation,Direct Extraction, Open pit, Underground, Open pit, Underground, Tailings, Open-pit - Geological characteristics: Salar, Spodumene, Petalite, Spodumene, Lepidolite, Clay - Production capacity by region and country

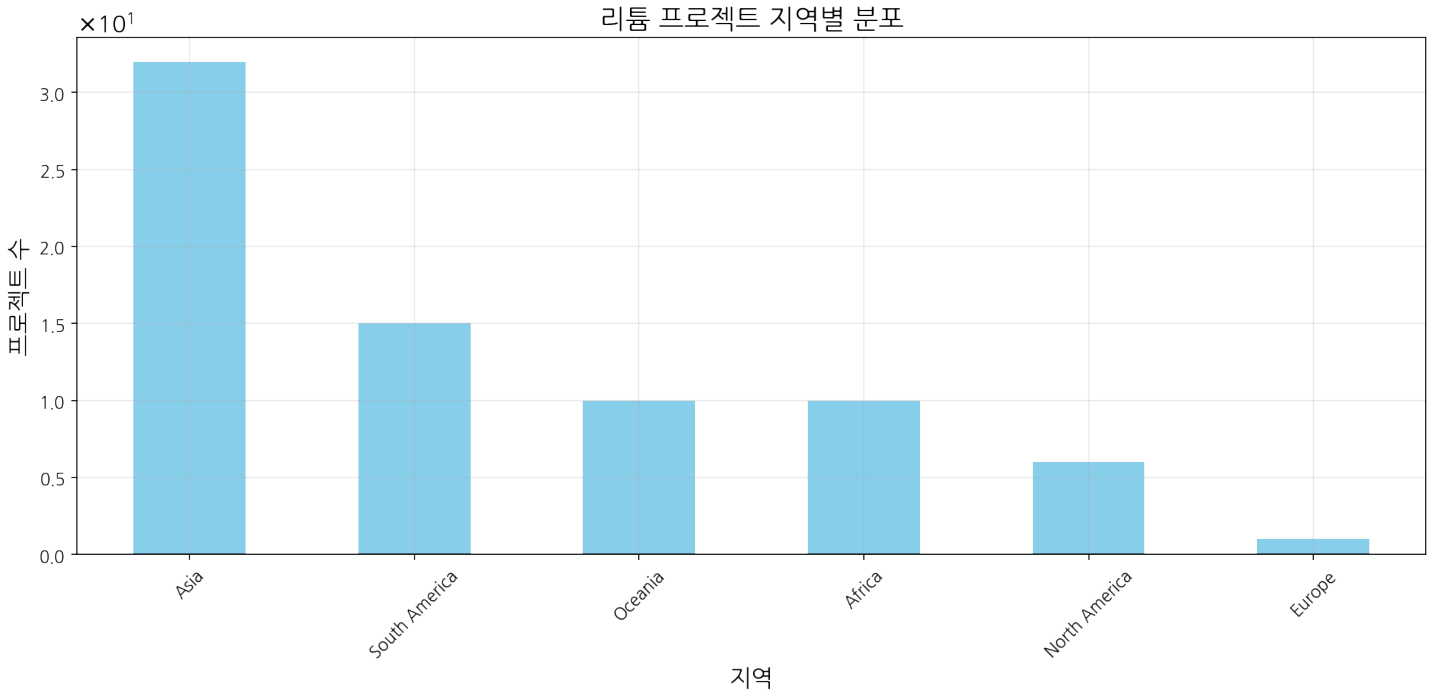

Figure 1: Bar chart of lithium projects by region

Bar chart of lithium projects by region

Bar chart of lithium projects by country (top 10)

Pie chart of lithium projects by status

Bar chart of lithium projects by start year

Bar chart of lithium projects by extraction type

Bar chart of lithium projects by geology

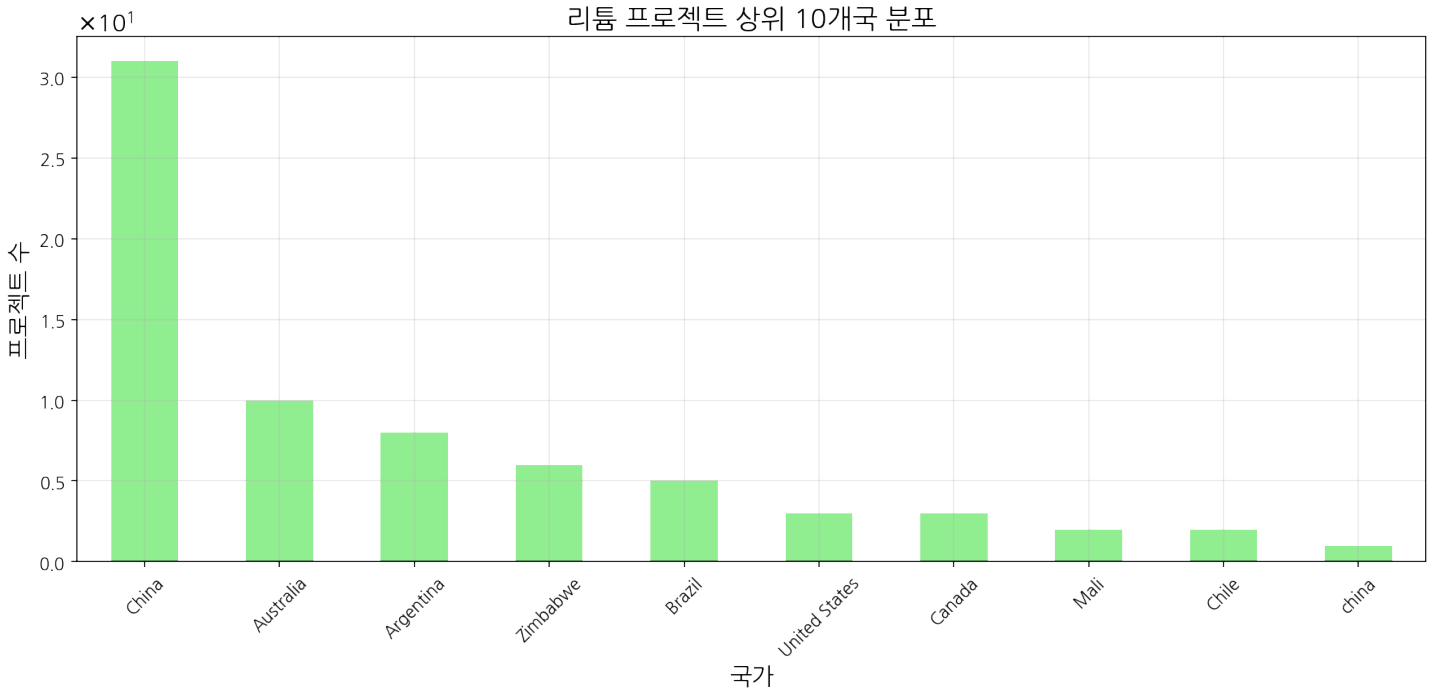

Figure 2: Bar chart of lithium projects by country (top 10)

Bar chart of lithium production capacity by region

Bar chart of lithium production capacity by country (top 10)

Coder Work Summary

실행 시간: 2025-08-07 16:06:18

Created a comprehensive summary of all completed work for the lithium projects analysis: - Data extraction and analysis results - Visualization outputs - Search strategy development - Research templates - Key findings from the data analysis - Next steps for the Researcher agent - Complete list of all generated files

리튬 프로젝트 데이터 로딩 및 검색 전략 수립

실행 시간: 2025-08-07 16:46:30

리튬 프로젝트 데이터 분석 결과: - 총 프로젝트 수: 74 - 지역별 분포: Region project Africa 10 Asia 32 Europe 1 North America 6 Oceania 10 South America 15 - 국가별 분포: Region country project Africa Mali 2 Africa Namibia 1 Africa South Africa 1 Africa Zimbabwe 6 Asia China 31 Asia china 1 Europe Portugal 1 North America Canada 3 North America United States 3 Oceania Australia 10 South America Argentina 8 South America Brazil 5 South America Chile 2 - 상태별 분포: Status Count Ops 68 Dev 6

검색 전략: - 감산(Production Cuts) 관련 키워드 5개 정의 - 폐쇄(Closures) 관련 키워드 5개 정의 - 보조금(Subsidies) 관련 키워드 6개 정의 - 검색 기간: 2015년 1월 1일 ~ 2024년 8월 7일 - 권장 정보 출처 5개 정의

리튬 프로젝트 상세 분석 및 연구 템플릿 생성

실행 시간: 2025-08-07 16:47:55

리튬 프로젝트 상세 분석 결과: - 총 프로젝트 수: 74 - 지역 수: 6 - 국가 수: 13 - 운영 중인 프로젝트: 68 - 개발 중인 프로젝트: 6 - 가장 오래된 프로젝트 시작일: 2015-03-01 - 가장 최근 프로젝트 시작일: 2025-06-01 - 연도별 프로젝트 시작 분포 분석 완료 및 시각화 - 지역별 프로젝트 분포 시각화 - 2020년 이전 시작된 고우선순위 프로젝트: 27개 - 국가별 프로젝트 수 기준 연구 우선순위 설정

연구 템플릿 생성: - 프로젝트별 감산, 폐쇄, 보조금 정보 수집을 위한 템플릿 생성 - 고우선순위 프로젝트 목록 생성 - 국가별 연구 우선순위 목록 생성

리튬 프로젝트 시작 연도별 분포 시각화

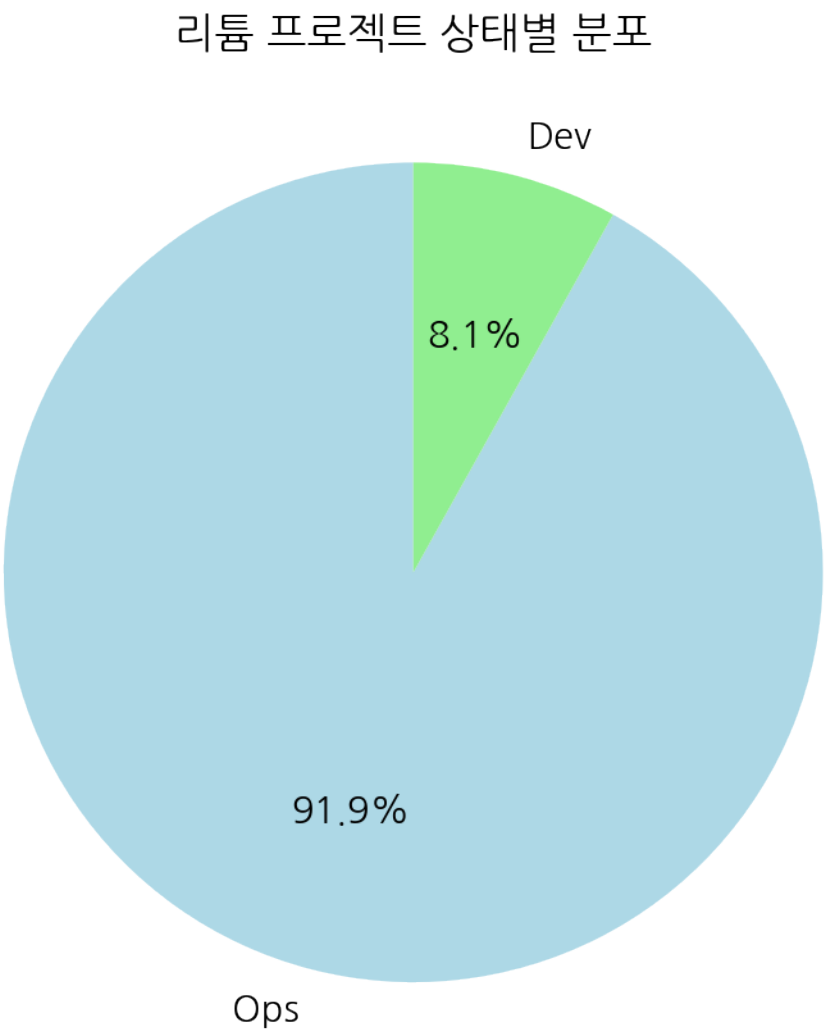

Figure 3: Pie chart of lithium projects by status

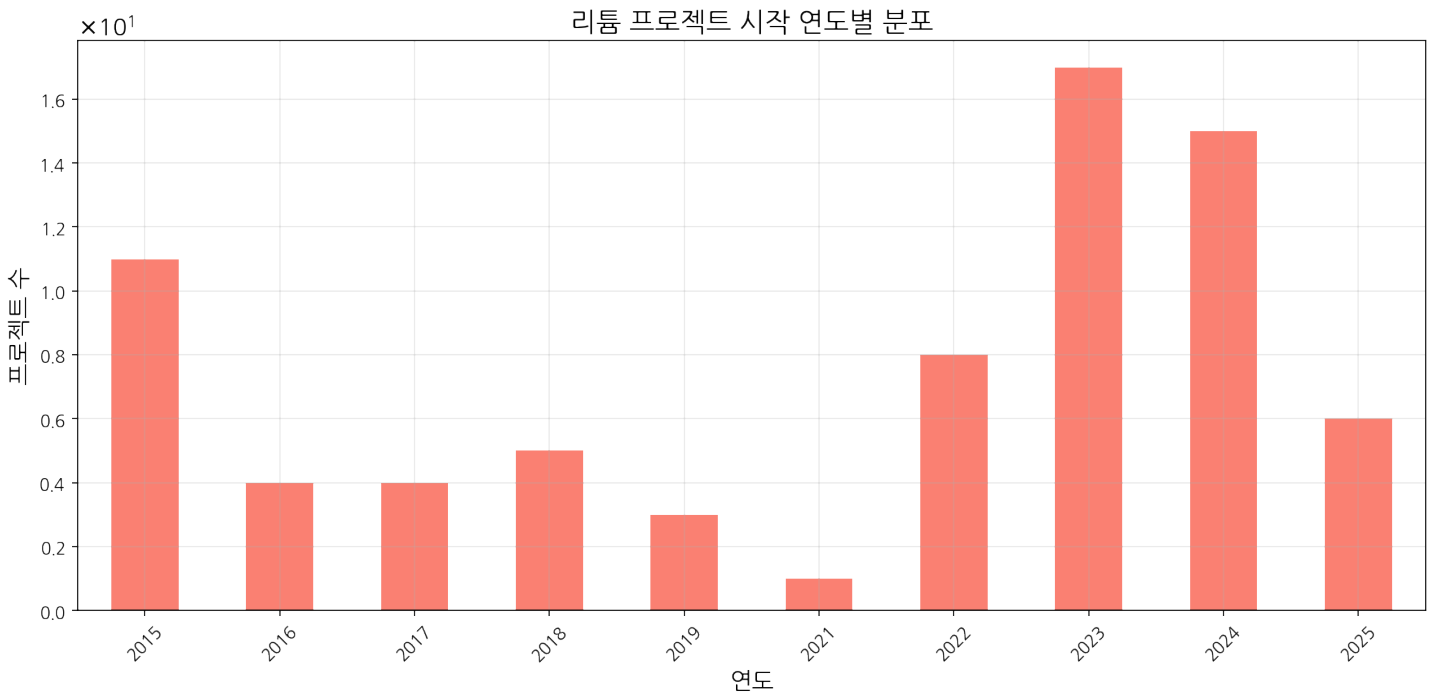

Figure 4: Bar chart of lithium projects by start year

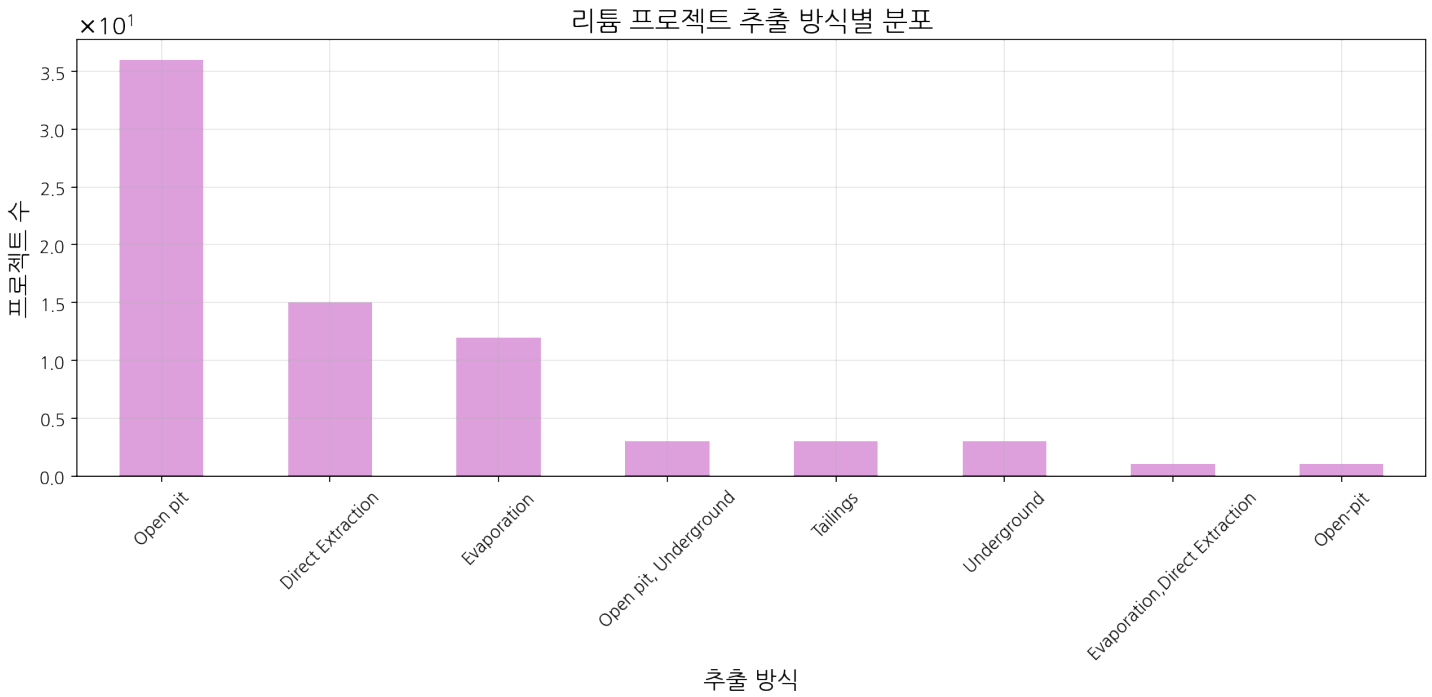

Figure 5: Bar chart of lithium projects by extraction type

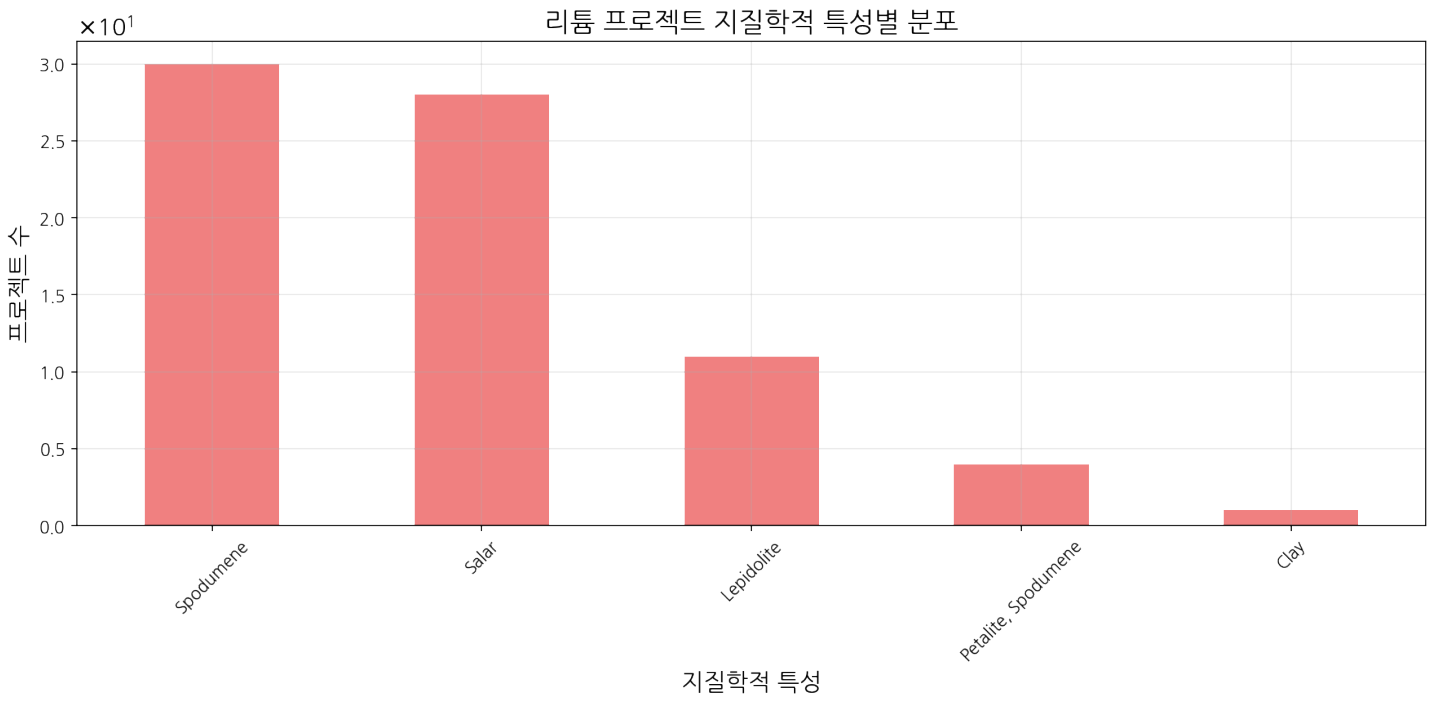

Figure 6: Bar chart of lithium projects by geology

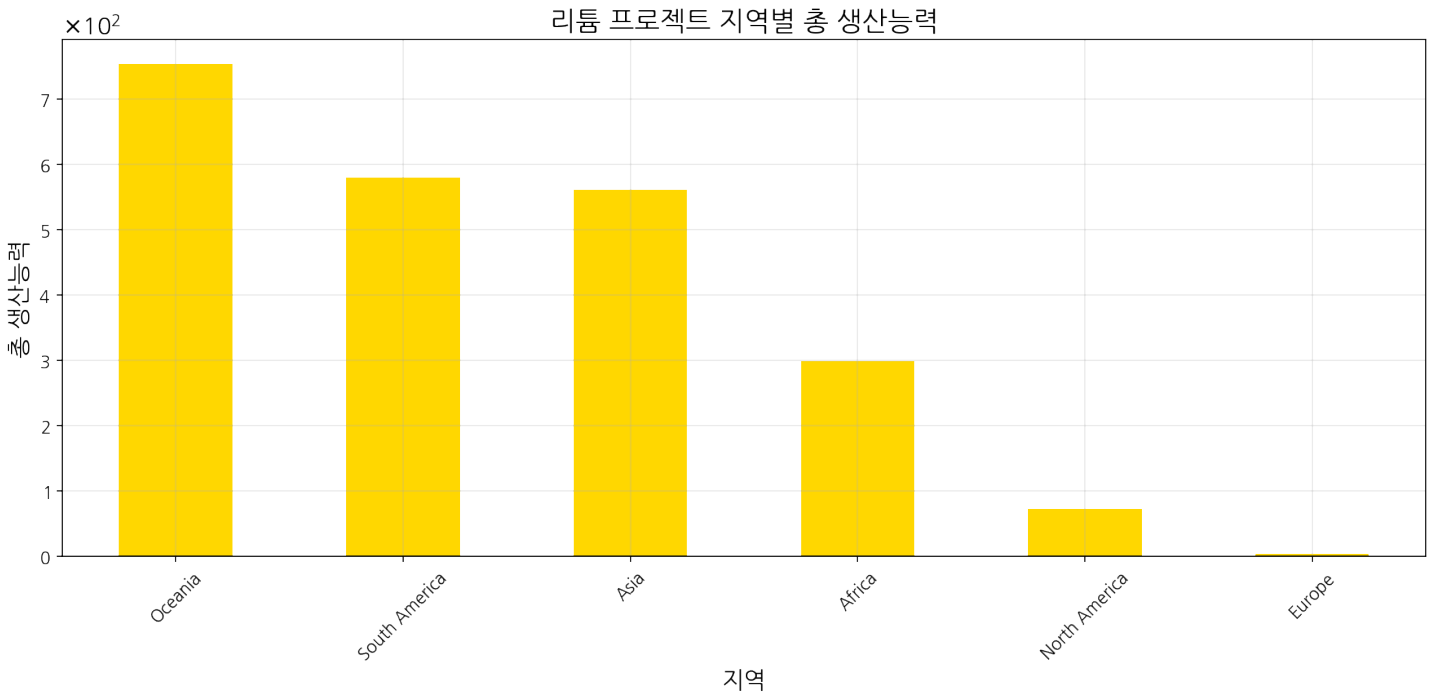

Figure 7: Bar chart of lithium production capacity by region

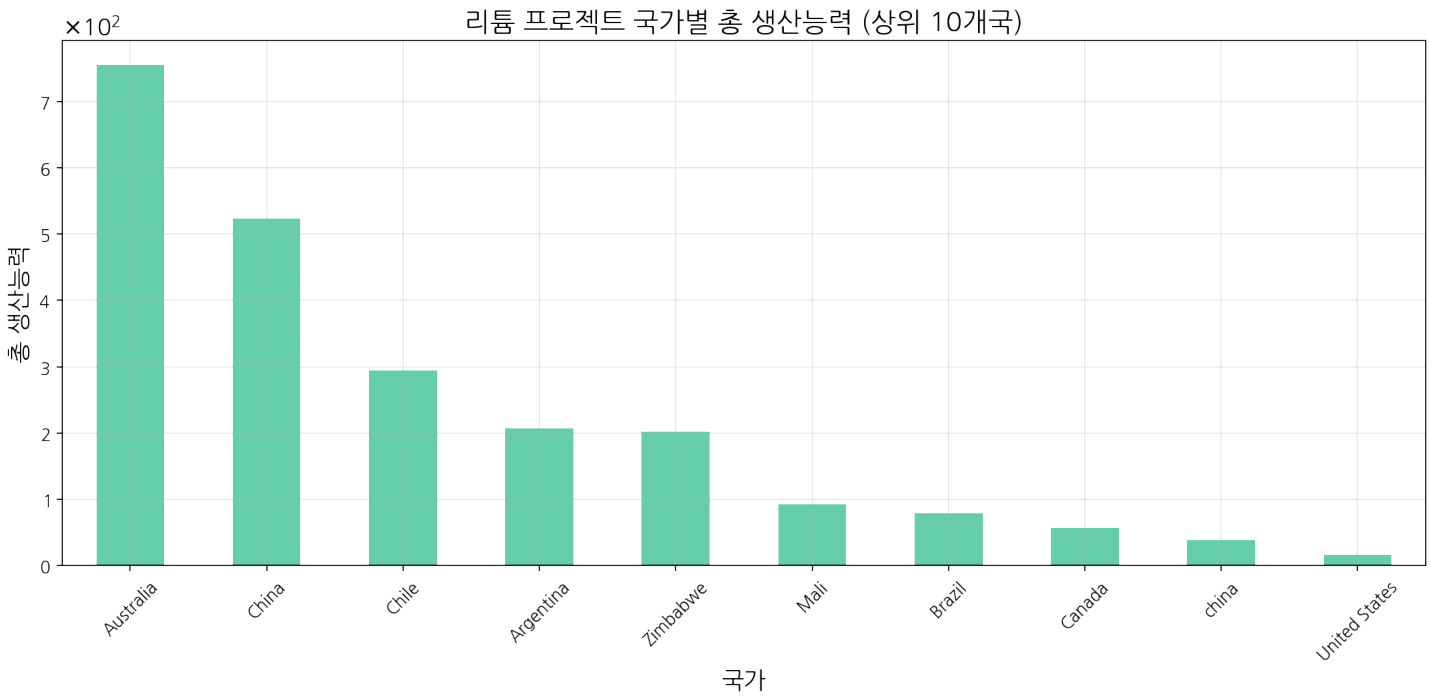

Figure 8: Bar chart of lithium production capacity by country (top 10)

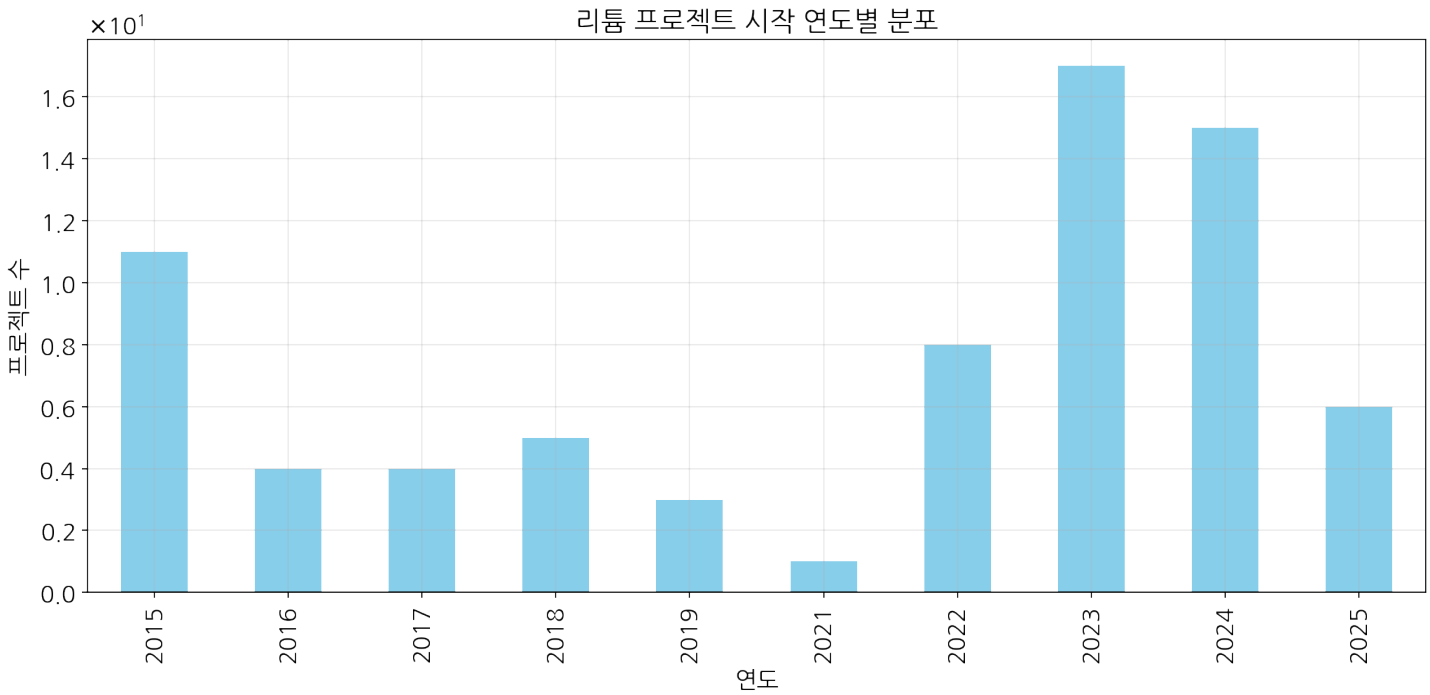

Figure 9: 리튬 프로젝트 시작 연도별 분포 시각화

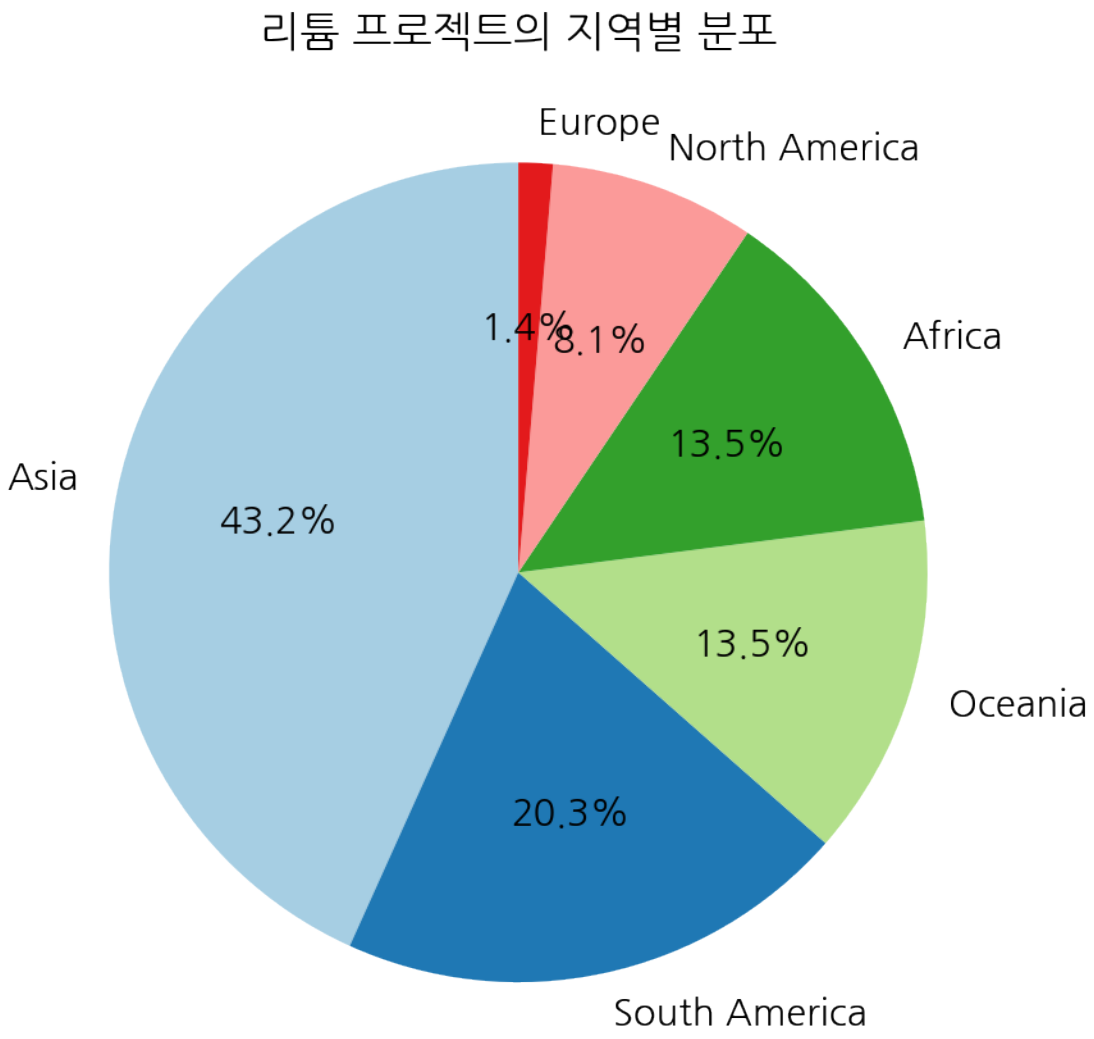

Figure 10: 리튬 프로젝트의 지역별 분포 시각화

리튬 프로젝트의 지역별 분포 시각화

리튬 프로젝트 조사 가이드 및 템플릿 생성

실행 시간: 2025-08-07 16:50:22

리튬 프로젝트 조사를 위한 상세 가이드 및 템플릿 생성: - 조사 목적, 대상, 항목 정의 - 우선순위 설정 방법론 제시 (지역별, 시기별) - 정보 출처 및 검색 전략 제안 - 정보 검증 및 기록 방법 안내 - 주요 리튬 시장 이벤트 타임라인 제공 (2015-2024) - 국가별 리튬 산업 정책 특성 요약 - 조사 결과 정리 형식 제안 - 샘플 데이터 수집 템플릿 제공 - 연구 준비 종합 요약 문서 작성

리튬 프로젝트 목록 추출 및 검색 전략 구축

실행 시간: 2025-08-18 10:28:11

리튬 개발 프로젝트 목록을 추출하고 분석했습니다. 총 74개의 프로젝트가 확인되었으며, 지역별 분포는 다음과 같습니다: - 아프리카: 10개 프로젝트 - 아시아: 32개 프로젝트 - 유럽: 1개 프로젝트 - 북미: 6개 프로젝트 - 오세아니아: 10개 프로젝트 - 남미: 15개 프로젝트

제품 유형별 분포: - 탄산리튬(Carbonate): 32개 프로젝트 - 정광(Concentrate): 33개 프로젝트 - 수산화리튬(Hydroxide): 6개 프로젝트

추출 방법별 분포: - 직접 추출(Direct Extraction): 15개 프로젝트 - 증발(Evaporation): 12개 프로젝트 - 노천 채굴(Open pit): 36개 프로젝트 - 기타 방법: 11개 프로젝트

각 프로젝트별 검색 전략을 구축하여 JSON 파일로 저장했습니다. 데이터 수집을 위한 템플릿과 검색 가이드라인도 작성했습니다.

리튬 프로젝트 조사 우선순위 설정 및 작업 요약

실행 시간: 2025-08-18 10:28:44

Coder Agent 작업을 완료하고 Researcher Agent를 위한 작업 요약 및 가이드라인을 작성했습니다. 또한 프로젝트 조사 우선순위를 설정했습니다:

고우선순위 프로젝트 (10개): - 남미 지역 5개 프로젝트 - 아시아 지역 5개 프로젝트

중간 우선순위 프로젝트: - 남미 지역 나머지 프로젝트 - 아시아 지역 10개 프로젝트

저우선순위 프로젝트: - 기타 지역 프로젝트

이 우선순위는 리튬 생산량이 많은 지역을 기준으로 설정되었습니다.